Hallucination Measurement Index Framework for Bilingual Malay-English Mental Health Chatbots

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HALLUCINATION MEASUREMENT INDEX FRAMEWORK FOR BILINGUAL MALAY-ENGLISH MENTAL HEALTH CHATBOTS

ABSTRACT

Large language models (LLMs) powering AI chatbots are increasingly deployed for mental health support, yet they remain prone to generating hallucinated content. Responses that are factually incorrect, clinically ungrounded, or contextually inappropriate present a risk to users seeking mental health support. This study proposes the Hallucination Measurement Index (HMI), a composite scoring index designed to measure the level of hallucination in AI chatbot responses within the mental health domain. The HMI is a student-designed composite index, representing the original contribution of this research. The HMI integrates five automated sub-metrics: Factual Consistency, Response Groundedness, Safety Compliance, Semantic Coherence, and Cultural Appropriateness, each operationalised using established NLP tools adapted for the Malaysian bilingual context. A custom bilingual scenario bank (MH-Hallu-MY) comprising 305 mental health prompts across 12 clinical categories is administered to three widely accessible LLM platforms powered by GPT-4 (OpenAI), Gemini Pro (Google DeepMind), and GPT-4 Turbo (Microsoft), generating 2,745 responses to evaluate and test the HMI index. The study employs non-parametric statistical tests including the Kruskal-Wallis H-test and Dunn's post-hoc analysis with Bonferroni correction to compare hallucination profiles across platforms. A Streamlit-based evaluation prototype operationalises the HMI for reproducible assessment. This research addresses a critical gap: the absence of a domain-specific, composite hallucination measurement index for mental health AI in Malaysia.

Keywords: Hallucination Measurement, AI Chatbots, Mental Health, Large Language Models, Malaysia.

HALLUCINATION MEASUREMENT INDEX FRAMEWORK FOR BILINGUAL MALAY-ENGLISH MENTAL HEALTH CHATBOTS

ABSTRAK

Model bahasa besar (LLM) yang menjanakan chatbot AI semakin banyak digunakan untuk sokongan kesihatan mental, namun mereka tetap terdedah kepada menjana kandungan halusinasi. Respons yang sebenarnya tidak betul, tidak berasas secara klinikal atau tidak sesuai mengikut konteks memberikan risiko kepada pengguna yang mendapatkan sokongan kesihatan mental. Kajian ini mencadangkan Indeks Pengukuran Halusinasi (HMI), indeks pemarkahan komposit yang direka untuk mengukur tahap halusinasi dalam respons AI chatbot dalam domain kesihatan mental. HMI ialah indeks komposit rekaan pelajar, mewakili sumbangan asal penyelidikan ini. HMI menyepadukan lima sub-metrik automatik: Ketekalan Fakta, Asas Tindak Balas, Pematuhan Keselamatan, Keselarasan Semantik dan Kesesuaian Budaya, masing-masing beroperasi menggunakan alat NLP yang telah disesuaikan untuk konteks dwibahasa Malaysia. Bank senario dwibahasa tersuai (MH-Hallu-MY) yang terdiri daripada 305 gesaan kesihatan mental merentas 12 kategori klinikal ditadbirkan kepada tiga platform LLM yang boleh diakses secara meluas yang dikuasakan oleh GPT-4 (OpenAI), Gemini Pro (Google DeepMind) dan GPT-4 Turbo (Microsoft), menjana 2,745 respons untuk menilai dan menguji indeks HMI. Kajian ini menggunakan ujian statistik bukan parametrik termasuk ujian Kruskal-Wallis H dan analisis post-hoc Dunn dengan pembetulan Bonferroni untuk membandingkan profil halusinasi merentas platform. Prototaip penilaian berasaskan Streamlit mengoperasikan HMI untuk penilaian boleh diterbitkan semula. Penyelidikan ini menangani jurang kritikal: ketiadaan indeks pengukuran halusinasi komposit khusus domain untuk AI kesihatan mental di Malaysia.

Kata kunci: Pengukuran Halusinasi, AI Chatbots, Kesihatan Mental, Model Bahasa Besar, Malaysia.

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LIST OF SYMBOLS AND ABBREVIATIONS

AI	:	Artificial Intelligence
API	:	Application Programming Interface
ASEAN	:	Association of Southeast Asian Nations
BERTScore	:	Bidirectional Encoder Representations from Transformers Score
BM	:	Bahasa Malaysia
CA	:	Cultural Appropriateness
DSM-5	:	Diagnostic and Statistical Manual of Mental Disorders, 5th Edition
EN	:	English
FC	:	Factual Consistency
GPT	:	Generative Pre-trained Transformer
HMI	:	Hallucination Measurement Index
ICD-11	:	International Classification of Diseases, 11th Revision
JSON	:	JavaScript Object Notation
LLM	:	Large Language Model
MH-Hallu-MY	:	Mental Health Hallucination Malaysia (scenario bank)
NHMS	:	National Health and Morbidity Survey
NLI	:	Natural Language Inference
NLP	:	Natural Language Processing
RAG	:	Retrieval-Augmented Generation
RG	:	Response Groundedness
RO	:	Research Objective
RQ	:	Research Question
SC	:	Safety Compliance
SD	:	Standard Deviation
SH	:	Semantic Coherence
VADER	:	Valence Aware Dictionary and sEntiment Reasoner
η^2	:	Eta-squared (effect size measure)
κ	:	Cohen’s Kappa (inter-rater reliability coefficient)
ρ	:	Spearman’s Rho (rank correlation coefficient)

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CHAPTER 1: INTRODUCTION

1.1 Background

The global mental health crisis has intensified over recent years. The World Health Organization reports that approximately one in eight people worldwide suffers from a mental health condition. In Malaysia, the 2023 National Health and Morbidity Survey (NHMS) revealed that 29.6% of adults experience mental health issues, nearly three times the rate from a decade ago. Consequently, there is an increasing demand for accessible mental health resources, and AI-powered chatbots are becoming valuable tools in addressing this need.

AI chatbots powered by large language models such as GPT-4 and Gemini show great promise in delivering empathetic and contextually suitable responses for mental health support (Abbasian et al., 2024). These systems employ transformer architectures to generate human-like text, offering ongoing mental health assistance without relying on the availability of human therapists. Nonetheless, the generative methods that make their responses smooth also introduce a major risk: hallucinations.

Hallucination in large language models (LLMs) happens when they generate content that is nonsensical, not aligned with the source, or contains factual inaccuracies (Ji et al., 2022). This problem has been observed across multiple natural language tasks, including summarisation, dialogue, and question answering (Ji et al., 2022; Rawte et al., 2023; Qi et al., 2024). In mental health chatbots, hallucinations can be particularly dangerous: they may lead to the creation of false therapeutic techniques, the suggestion of non-existent medications, the provision of unsafe clinical advice, or the failure to recognize urgent crisis situations requiring professional intervention.

While research on detecting and reducing hallucinations in general-purpose large language models (LLMs) is increasing (Luo et al., 2024; Abdelghafour et al., 2024), there remains a significant gap in creating domain-specific frameworks for assessing hallucinations in mental health contexts. Existing benchmarks like HaluEval (Li et al., 2023), TruthfulQA (Lin et al., 2022), and MedHallu (2025) provide useful benchmarks but do not specifically address the unique challenges of therapeutic conversations, particularly in multilingual and culturally diverse settings like Malaysia (Chataigner et al., 2024; Kang et al., 2024).

1.2 Problem Statement

While AI-powered mental health chatbots are becoming more common in Malaysia and globally, existing evaluation is limited by the absence of a standardized, domain-specific method to measure hallucination levels in their responses. Current evaluation methods have three main limitations. First, broad benchmarks like HaluEval (Li et al., 2023) and TruthfulQA (Lin et al., 2022) do not focus on clinical sensitivity, safety, or domain-specific knowledge that are essential for mental health contexts. Second, existing methods often depend on a single metric, such as factual accuracy or semantic similarity, rather than providing a thorough, multi-dimensional evaluation that captures different hallucination types relevant to therapeutic conversations (Das et al., 2022; Dziri et al., 2022). Third, most hallucination research is conducted in English, neglecting the multilingual and cultural considerations important for countries like Malaysia, where AI chatbots must operate in both English and Bahasa Malaysia (Chataigner et al., 2024).

A further critical gap concerns cross-platform empirical evidence. Although commercial LLM platforms exhibit documented differences in hallucination tendency on general-purpose benchmarks (Vectara, 2024; Shan et al., 2024), no standardized, reproducible comparison of these platforms has been conducted within the Malaysian mental health domain. This absence has two consequences: clinicians, policymakers, and AI procurement decision-makers in Malaysia currently lack empirical evidence to guide platform selection for sensitive clinical conversations, and any new domain-specific hallucination index requires a principled comparative test to demonstrate that it can discriminate between platforms in this context. The present study therefore positions the three-platform comparison (GPT-4, Gemini Pro, GPT-4 Turbo) as both an empirical contribution to Malaysian digital health policy and a construct-validity test of the proposed HMI index.

In Malaysia, the absence of a formal framework leads to AI chatbots for mental health support being used without rigorous quality checks for accuracy, clinical validity, or cultural appropriateness. This gap creates risks for vulnerable users, who may receive false or hallucinated clinical advice without a clear way to recognize or assess these errors.

1.3 Research Objectives

The general aim of this study is to develop and validate a domain-specific, bilingual composite index that reliably measures hallucination in AI mental-health chatbot responses within the Malaysian English-Malay context.

This study aims to achieve the following three research objectives:

RO1: To design a composite Hallucination Measurement Index (HMI) model that integrates multiple automated NLP sub-metrics for evaluating hallucination in mental health chatbot responses.

RO2: To evaluate and compare the hallucination profiles of three widely accessible LLM platforms using the HMI across bilingual mental health scenarios.

RO3: To develop a functional evaluation prototype that operationalizes the HMI for reproducible hallucination assessment of mental health chatbot responses.

1.4 Research Questions

The following research questions guide this study:

RQ1: What combination of automated NLP sub-metrics can effectively capture the multi-dimensional nature of hallucination in mental health chatbot responses?

RQ2: How do the hallucination profiles of the three selected LLM platforms differ when evaluated using the HMI across bilingual mental health scenarios?

RQ3: How can the HMI be operationalised into a functional prototype for reproducible hallucination evaluation?

1.5 Research Scope

This study examines hallucinations in text responses from three publicly available LLM platforms: GPT-4 (OpenAI), Gemini Pro (Google DeepMind), and GPT-4 Turbo (Microsoft). All are accessed via their free-tier web interfaces to ensure reproducibility. The research focuses on mental health by using a custom bilingual scenario bank (MH-Hallu-MY) featuring 305 prompts in 12 clinical categories in both English and Bahasa Malaysia. This quantitative, automated study involves no human participants and does not create chatbots. The primary outputs are a measurement model (HMI) and an evaluation prototype, rather than a new chatbot system.

1.6 Significance of Study

This study makes three key contributions. First, it addresses a gap identified by Abbasian et al. (2024) and Aljamaan et al. (2024) regarding the need for domain-specific hallucination assessment in healthcare AI by creating the first composite index specifically designed for evaluating mental health chatbot responses. Second, it expands the scope of hallucination research beyond primarily English-language studies (Chataigner et al., 2024; Kang et al., 2024) by incorporating bilingual evaluation in

English and Bahasa Malaysia, reflecting Malaysia's multilingual AI landscape. Third, the open-source evaluation tool enables practitioners, regulators, and researchers to accurately assess chatbot safety before deployment, supporting responsible AI governance within Malaysia’s healthcare sector.

The following diagram summarises the overall research flow from the identified problem to the proposed HMI solution.

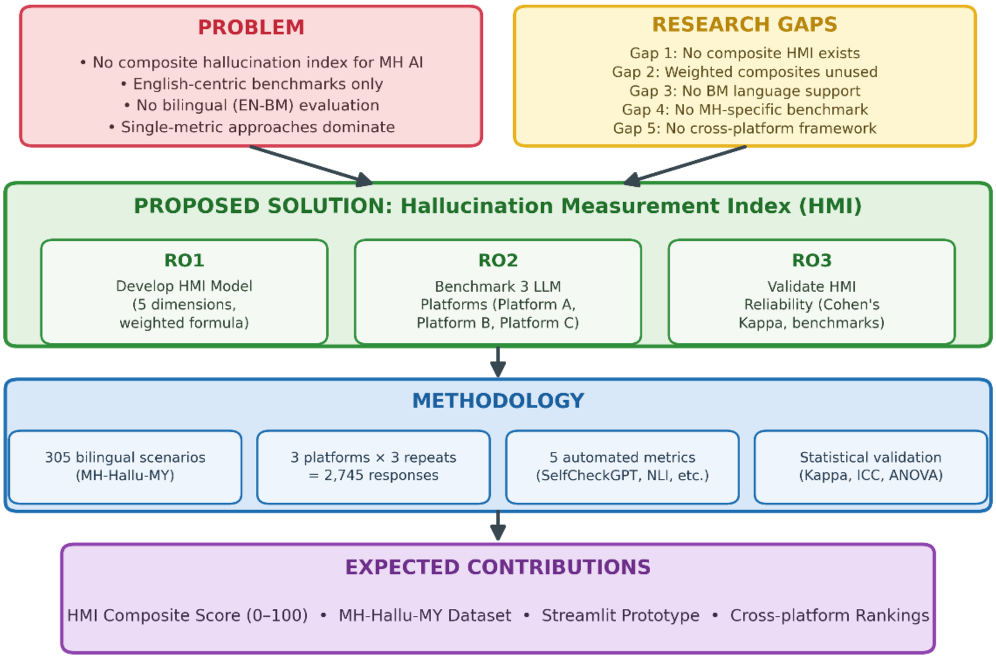

Figure 1.1: Research Overview – From Problem to Solution

1.7 Definition of Key Terms

Hallucination occurs when an LLM produces content that is nonsensical, not true to the source, or factually incorrect (Ji et al., 2022). In this context, it includes factual errors, clinical misinformation, safety issues, and culturally sensitive material.

Large Language Model (LLM): A neural network trained on extensive text datasets with transformer architecture that produces human-like text (Rawte et al., 2023).

Examples include OpenAI's GPT-4, Google DeepMind's Gemini Pro, and Microsoft's GPT-4 Turbo deployment.

Hallucination Measurement Index (HMI): This study introduces a composite scoring model that incorporates five automated sub-metrics to generate a normalized hallucination score (0-100) for chatbot responses within the mental health domain.

Mental Health Chatbot: An AI-driven conversational agent created to offer mental health support, information, or therapy through text-based interactions (Abbasian et al., 2024).

SelfCheckGPT: A zero-resource black-box hallucination detection method that measures factual consistency by comparing multiple sampled responses from the same model (Manakul et al., 2023).

MH-Hallu-MY: The custom bilingual mental health scenario bank developed for this study, comprising 305 prompts across 12 clinical categories in English and Bahasa Malaysia.

1.8 Organization of the Thesis

This thesis is structured into five distinct chapters. Chapter 1 offers an introduction that encompasses the background, problem statement, research objectives and questions, scope, and significance. Chapter 2 provides an in-depth literature review that addresses AI applications in mental health, taxonomy and detection of hallucinations, existing measurement frameworks, and the Malaysian context. Chapter 3 outlines the research methodology, which includes the design of the HMI index, the experimental protocol, and specifications for the prototype. Chapter 4 showcases the results and analysis stemming from the hallucination evaluation experiment. Chapter 5 reflects on the findings, discusses their implications, acknowledges limitations, and suggests avenues for future research.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

This chapter presents an extensive review of the literature that supports the creation of the Hallucination Measurement Index (HMI). The review is structured thematically, starting with the impact of AI on mental health services, moving through the classification of hallucinations, means of detection, and current benchmarks, and finishing with a discussion of research gaps that this study aims to address. All referenced works are sourced from the curated collection of papers provided by the research supervisor, along with essential foundational literature in the field. Figure 2.1 depicts the chronological progression of significant research advancements that have led to this study.

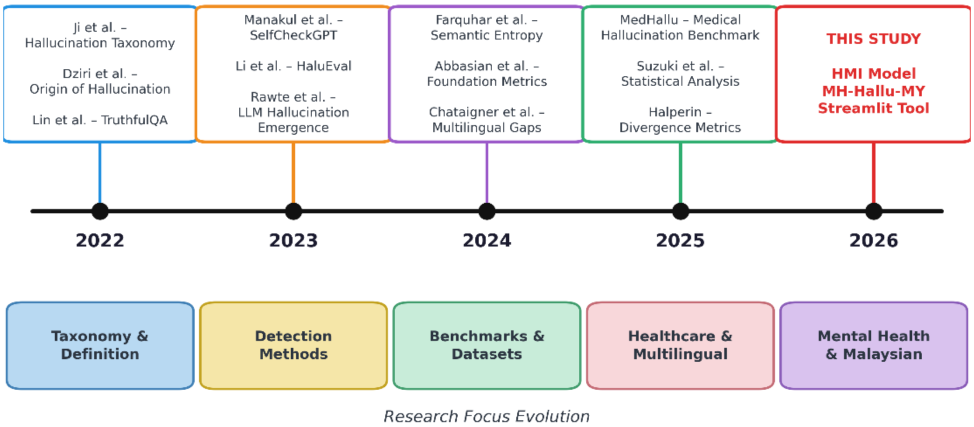

Figure 2.1: Evolution of Hallucination Research Leading to the HMI

2.2 Artificial Intelligence in Mental Health Care

The integration of artificial intelligence into mental health care has progressed significantly over the last ten years, fueled by advancements in natural language processing and an increasing need for scalable mental health solutions. Abbasian et al. (2024) introduced foundational metrics aimed at assessing the effectiveness of healthcare dialogues enhanced by generative AI, demonstrating that AI chatbots can produce clinically significant results when appropriately assessed. Their framework, published in npj Digital Medicine, outlined five essential dimensions of the quality of healthcare conversations, setting a methodological standard for multi-dimensional evaluation that this study aims to expand upon in the specific area of measuring hallucinations.

The use of AI chatbots for mental health assistance has been recorded across various platforms and demographics. Nevertheless, Aljamaan et al. (2024) raised an important issue in their examination of AI chatbot medical hallucination: they created the References Hallucination Score and evaluated numerous chatbot systems, discovering that all the commercial chatbots assessed demonstrated notable occurrences of hallucinated medical references. This result highlights the necessity of establishing strong measurement tools specifically designed for healthcare AI applications.

2.3 Mental Health Chatbots

Mental health chatbots are a specific use of conversational AI crafted to offer therapeutic support, educational resources, or crisis intervention via text-based conversations. Gatti et al. (2024) explored this area through the perspective of runtime verification in their study RV4Chatbot, questioning whether chatbots should be permitted to "dream" - a metaphor for hallucinations in therapeutic settings. Their research revealed that the outputs generated by chatbots need ongoing oversight to ensure clinical safety, a guideline that directly influences the Safety Compliance aspect of the Human-Machine Interaction (HMI).

Priola (2024) explored the issue of hallucinations in healthcare chatbots by integrating Retrieval-Augmented Generation (RAG) with a new metric known as NMISS (Normalized Medical Information Semantic Similarity), specifically within the scope of Italian healthcare LLM chatbots. This research illustrated that grounding mechanisms tailored to the domain can significantly decrease the occurrence of hallucinations, providing proof that the Response Groundedness aspect of the HMI is both quantifiable and capable of improvement.

2.4 Hallucination in Large Language Models (LLM)

2.4.1 Definition and Taxonomy of Hallucination

Ji et al. (2022) carried out a groundbreaking survey on hallucinations in natural language generation, defining hallucination as "content that is either nonsensical or not aligned with the original source." Their detailed taxonomy distinguished between intrinsic hallucination, which contradicts the source, and extrinsic hallucination, which cannot be verified against it. This classification framework has gained widespread acceptance. Published in ACM Computing Surveys, this survey remains a key reference for research on hallucination.

Rawte et al. (2023) expanded this classification system specifically for large language models in their paper "The Troubling Emergence of Hallucination in Large Language Models," presented at EMNLP 2023. They identified new hallucination categories like factual fabrication, context deviation, and instruction non-compliance, providing a more detailed classification relevant to conversational AI. Their quantification framework contributed to the multi-dimensional approach used in the HMI.

Qi and colleagues (2024) surveyed the evolution of hallucination assessment in Natural Language Generation (NLG), noting a shift from simple n-gram matching to more sophisticated model-driven evaluations. They categorized evaluation methods into three groups: reference-based, reference-free, and LLM-based, finding that combined metrics outperform single-metric assessments. This finding supports HMI’s design, which uses a multi-metric composite approach.

2.4.2 Causes of Hallucination in LLMs

Dziri et al. (2022) investigated the origins of hallucinations in conversational models, focusing on whether they stem from the training data or the model architecture. Their research on knowledge-grounded dialogue systems showed that both factors are involved, with the quality of training data having a greater impact. Presented at NAACL 2022, this study introduced a metric to measure hallucinations in dialogues, enhancing the Factual Consistency aspect of the HMI.

Suzuki et al. (2025) offered a theoretical perspective, proposing that hallucinations are "unavoidable yet statistically insignificant" in properly calibrated systems. However, in mental health contexts, even rare but severe hallucinations,such as dangerous clinical recommendations,can have serious consequences. This necessitates including a Safety Compliance component in the HMI with significant emphasis.

2.4.3 Hallucination in Dialogue Systems

Das et al. (2022) conducted a detailed study of fact hallucination patterns in dialogue systems. They identified eight distinct modes of hallucination specific to knowledge-grounded conversations and developed the FADE (FActual Dialogue Hallucination Detection) dataset for comprehensive evaluation. Their entity-level hallucination detection approach provides a foundation for detailed assessment, which the HMI adapts for mental health applications.

Lotfi et al. (2022) introduced ConsisTest, a benchmark designed to evaluate factual consistency in generative conversational systems. Their comprehensive evaluation method, merging symbolic and sub-symbolic techniques, showed that traditional metrics such as BLEU (Bilingual Evaluation Understudy) and perplexity are insufficient for detecting factual inaccuracies. Instead, targeted evaluations are more effective in identifying these discrepancies. This finding aligns with the HMI's preference for using

NLI-based groundedness verification rather than relying solely on text similarity measures.

Furumai et al. (2023) employed Graph Neural Networks (GNNs) to detect hallucinations in dialogues, demonstrating that the relationships between statements and knowledge sources can be effectively modeled for automated identification. This study validates the feasibility of automated hallucination detection within conversational environments.

2.5 Hallucination Detection and Measurement Methods

2.5.1 Automated Detection Methods

Farquhar et al. (2024) published a groundbreaking study in Nature demonstrating that semantic entropy, a measure of uncertainty in LLM outputs, can effectively detect hallucinations. Their method analyzes the distribution of meanings across multiple samples, providing a theoretical foundation for the SelfCheckGPT approach used in the HMI's Factual Consistency dimension.

Manakul et al. (2023) introduced SelfCheckGPT, a black-box hallucination detection technique that doesn't need external databases or internal model details. It samples multiple responses to the same prompt and evaluates their consistency, providing a practical, scalable solution suited for testing commercial chatbots with only API or web access. This approach is core to the HMI's Factual Consistency sub-metric.

Halperin (2025) proposed metrics that measure the semantic divergence between prompts and responses to identify faithfulness hallucinations, assessing the semantic distance between user queries and chatbot responses. This advances the Semantic Coherence aspect of HMI.

2.5.2 Benchmark Hallucination Datasets

Li et al. (2023) developed HaluEval, a comprehensive benchmark for hallucination assessment with 35,000 samples across question answering, dialogue, and summarization tasks. This dataset has been used to evaluate different commercial LLM platforms, such as the Gemini model family (Shan et al., 2024). It is publicly available on HuggingFace under the username pminervini, licensed under Apache 2.0.

Lin et al. (2022) introduced TruthfulQA, a benchmark assessing how models generate inaccuracies similar to common human misconceptions. Launched at ACL 2022, it contains 817 questions divided into 38 categories, such as health topics. The dataset is available on HuggingFace (truthfulqa/truthful_qa, Apache 2.0).

MedHallu (2025) introduced the first comprehensive benchmark aimed at detecting medical hallucinations in LLMs. The dataset can be accessed on HuggingFace (UTAustin-AIHealth/MedHallu). Although MedHallu covers general medicine, it does not specifically focus on mental health or conversational contexts.

2.5.3 Existing Measurement Models and Indices

Raghava (2024) created a classification system for hallucinations that uses a new weighted metric, showing that various hallucination types need different measurement approaches. It also finds that a combined weighted score outperforms individual metrics. This supports the HMI's weighted multidimensional framework with empirical evidence.

Abbasian et al. (2024) presented Foundation Metrics in npj Digital Medicine to evaluate healthcare conversation effectiveness. While this framework is the most similar existing work to the HMI, it considers overall conversation quality without specifically targeting hallucinations. The HMI extends this by focusing exclusively on measuring hallucinations.

Thomas et al. (2024) conducted a cost-effectiveness analysis of various hallucination detection systems, evaluating the trade-offs between computational cost, detection accuracy, and ease of deployment across twelve methods. They found that combining multiple lightweight metrics into composite scoring can deliver accuracy comparable to expensive single-model approaches, but with much less computational demand.

In 2024, Rawte et al. introduced FACTOID, a framework for factual entailment in hallucination detection that combines NLI (Natural Language Inference)-based verification with entity-level checks, achieving top results on multiple benchmarks. Luo et al. also in 2024, thoroughly examined hallucination detection and mitigation strategies, highlighting the available tools and noting that no existing framework fully unifies detection, measurement, and domain-specific evaluation in one index. These findings support the HMI's design as a cost-effective, multi-metric framework. Figure 2.2 compares these methods with the HMI across nine key evaluation features.

Comparison of Existing Hallucination Evaluation Approaches

Feature	Ji et al. (2022)	Manakul et al. (2023)	Raghava (2024)	Abbasian et al. (2024)	MedHallu (2025)	HMI (This Study)
Composite Index	No	No	Yes	Yes	No	Yes
Mental Health Focus	No	No	No	Partial	No	Yes
Multi-Dimensional	No	No	Yes	Yes	No	Yes
Bilingual (EN-BM)	No	No	No	No	No	Yes
Automated Scoring	Partial	Yes	Yes	Partial	Yes	Yes
Cross-Platform	No	No	No	Partial	No	Yes
Open-Source Tool	No	Partial	No	No	Yes	Yes
Cultural Context	No	No	No	No	No	Yes
Safety Compliance	No	No	No	Partial	Partial	Yes

Legend: **Yes = Fully Supported** **Partial = Partially Supported** **No = Not Supported**

The HMI is the only framework that addresses all nine evaluation features simultaneously.

Figure 2.2: Comparison of Existing Hallucination Evaluation Approaches

Across the reviewed literature, five recurring dimensions emerge as the basis for evaluating hallucination in conversational AI: factual consistency (Manakul et al., 2023), response groundedness (Priola, 2024), safety compliance (Aljamaan et al., 2024), semantic coherence (Halperin, 2025), and cultural appropriateness (Chataigner et al., 2024). These five dimensions collectively motivate the composite design of the HMI; each is operationalised and explained in detail in Section 3.2.

2.6 Hallucination Evaluation Datasets for NLP and Healthcare

2.6.1 Mental Health and Dialogue Datasets

Several publicly accessible datasets offer ground truth data for assessing mental health conversations. MentalChat16K (ShenLab, 2025) includes over 16,000 synthetic counseling dialogues with verified clinical content, available on HuggingFace under an MIT license. ESConv (Liu et al., 2021) features 1,300 emotional support conversations comprising 31,410 utterances and eight support strategies, published at ACL 2021 and accessible via HuggingFace (thu-coai/esconv, CC-BY-NC-4.0). Counsel Chat (Bertagnolli, 2020) contains more than 1,000 real question-answer pairs from professional therapists, available on HuggingFace (nbertagnolli/counsel-chat). These datasets serve as the knowledge base for the HMI's Response Groundedness dimension.

2.6.2 Hallucination dataset Benchmarks

The SelfCheckGPT WikiBio dataset (Manakul et al., 2023) contains human-annotated labels for hallucinations in GPT-3 generated biographical passages and is accessible on HuggingFace (potsawee/wiki_bio_gpt3_hallucination, CC-BY-SA-3.0, over 29,500 downloads). This dataset is used as the calibration reference for the HMI's Factual Consistency dimension. Additionally, the Hallucinations Leaderboard (Vectara, 2024) on HuggingFace compiles benchmark results from multiple models, with over 58,400 downloads, allowing comparison of HMI chatbot rankings against broader community benchmarks.

2.7 Hallucination in Healthcare and Mental Health AI

Priola (2024) demonstrated that using RAG combined with NMISS reduces hallucinations in Italian healthcare chatbots. Aljamaan et al. (2024) concentrated on medical hallucinations and developed the References Hallucination Score. Nambiar and Sreedevi (2023) proposed consensus strategies across multiple chatbots to reduce AI hallucinations. All these strategies rely on the idea that hallucinated responses tend to have lower inter-response consistency.

Liu (2024) surveyed the types and causes of hallucinations in large language models, noting that the risks are heightened in high-stakes domains such as healthcare, finance, and law. The study emphasized that healthcare contexts demand a more solid factual foundation because patients might follow AI suggestions without professional verification. Aljamaan et al. (2024) supported this view with their Reference Hallucination Score, which evaluated six commercial AI chatbots on medical prompts and found substantial hallucination in their cited references (up to 61.6% for reference relevancy), underscoring that even widely used models hallucinate in medical contexts. These results highlight the importance of developing measurement tools tailored to healthcare AI, rather than relying on general metrics.

In mental health, hallucinations can have serious consequences. Muqtadir et al. (2024) found that AI chatbots often mix diagnostic rules, invent therapy methods, or give harmful advice, especially on issues like suicidal thoughts. Boucher et al. (2021) reviewed AI chatbots in digital mental health interventions, noting that while these tools can support engagement and education, their reliability in critical cases requires careful human oversight. Together, these studies show that AI in mental health needs evaluation tools that address both accuracy and clinical safety, a need that the HMI seeks to meet.

2.8 Malaysian Context and Bilingual Challenges

Chataigner et al. (2024) pointed out discrepancies in multilingual hallucinations in LLMs, noting that these models tend to hallucinate more frequently in non-English languages. They also found that detection metrics are less reliable across different languages. Kang et al. (2024) conducted a detailed comparison of hallucination detection metrics for multilingual outputs, discovering that metrics effective for English do not perform well in other languages. These findings directly impact the HMI's bilingual evaluation, which uses the MH-Hallu-MY scenario bank with 150 prompts in Bahasa Malaysia and 155 in English.

Malaysia's cultural context regarding mental health requires careful awareness of local specifics. Elements like religious influences on coping, societal stigma, and the structure of the Malaysian healthcare system call for particular considerations that standard metrics can't fully capture. The HMI's Cultural Appropriateness dimension addresses this by employing a custom rule-based assessment.

Kang et al. (2024) examined hallucination detection across languages, finding that detection-metric accuracy drops markedly for non-English generation. Although Bahasa Malaysia is not strictly low-resource, it is mid-resource, with commercially used LLMs having much less training data than English. Chataigner et al. (2024) and Kang et al. (2024) showed that hallucination behaviour and detection accuracy differ across languages, justifying the HMI's approach to evaluating both English and Bahasa Malaysia prompts within a unified scoring system.

Furthermore, Chataigner et al. (2024) highlighted the importance of culturally and linguistically aware AI assessment. They pointed out that benchmarks based on Western standards often underestimate hallucination rates in non-Western contexts because of the absence of culturally relevant knowledge bases. In the Malaysian mental health sector, this can lead to chatbot outputs that recommend inappropriate coping strategies, misrepresent local mental health services, or miss culturally specific signs of distress. Gosmar and Dahl (2025) demonstrated that orchestrating multiple specialised AI agents can help mitigate hallucinations in conversational AI systems. This approach is exemplified in the MH-Hallu-MY dataset, which contains 150 Bahasa Malaysia scenarios tailored to Malaysian cultural contexts.

2.9 Research Gap

The literature review identifies six key research gaps that collectively motivate this study:

Gap 1: Currently, there is no unified composite hallucination index to assess mental health AI. While Ji et al. (2022) and Rawte et al. (2023) created detailed hallucination taxonomies, and Raghava (2024) demonstrated that weighted composite metrics are more effective, no study has integrated these approaches into a single measurement tool tailored for mental health chatbots. Existing frameworks typically focus on either general natural language generation (Ji et al., 2022), broad healthcare conversations (Abbasian et al., 2024), or single-dimension detection (Manakul et al., 2023). The HMI fills this gap by merging five related dimensions into one comprehensive score from 0 to 100.

Gap 2: Many evaluations depend on a single metric, despite evidence that weighted composites are more effective. Raghava (2024) demonstrated through empirical data that multi-metric weighted strategies can detect hallucination patterns overlooked by single metrics; still, most current tools rely on only one detection method. Thomas et al. (2024) confirmed that composite approaches offer better cost-effectiveness. The HMI addresses this gap with its weighted formula, which considers five dimensions.

Gap 3: Most current benchmarks emphasize English-speaking environments. Studies by Chataigner et al. (2024) and Kang et al. (2024) found that hallucination rates increase and detection becomes less accurate in non-English languages. Kang et al. (2024) noted similar trends in moderately-resourced languages. No benchmark currently includes scenarios in Bahasa Malaysia. The MH-Hallu-MY benchmark addresses this gap with 150 prompts in Bahasa Malaysia and 155 prompts in English focused on mental health.

Gap 4: Current healthcare benchmarks tend to overlook mental health specifically. While MedHallu (2025) and Priola (2024) address general medicine and Italian healthcare respectively, they do not focus on hallucination patterns unique to mental health, such as false therapeutic techniques, inappropriate crisis advice, or culturally mismatched coping strategies (Muqtadir et al., 2024). The MH-Hallu-MY scenarios are specifically designed around categories relevant to mental health.

Gap 5: There is no standardized framework for comparing across different platforms. Shan et al. (2024) benchmarked the Gemini and Kimi LLM platforms but used inconsistent metrics that are not universally reproducible. Likewise, Qi et al. (2024) observed that hallucination evaluation methods vary greatly across studies, with no clear, unified scoring approach. The HMI offers a reproducible, standardized framework for evaluating any chatbot available via web interface or API.

Gap 6: There is currently no open-source evaluation tool available. Although hallucination research has increased, Thomas et al. (2024) and Abbasian et al. (2024) pointed out that practical, deployable evaluation tools are still scarce. The HMI prototype, built as a Streamlit dashboard using open-source NLP libraries, provides a reusable resource for researchers and practitioners.

The following chart summarises the identified research gaps and shows how the HMI addresses both coverage and impact gaps in mental-health chatbot hallucination evaluation.

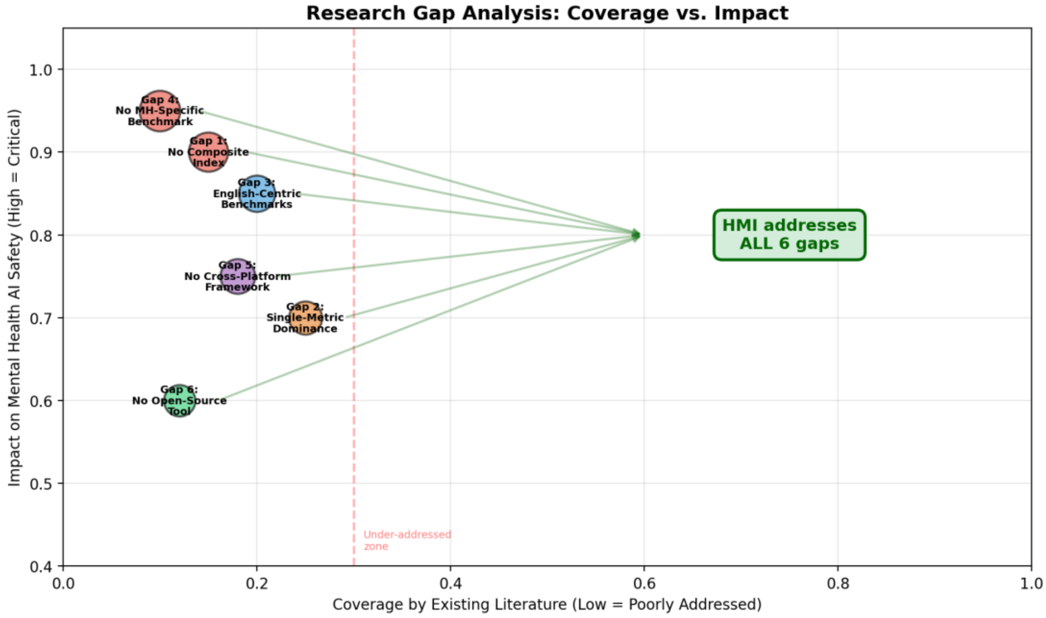

Figure 2.3: Research Gaps Analysis - Coverage vs. Impact of Six Identified Gaps in Mental Health Chatbot Hallucination Research

2.10 Conceptual Framework

The conceptual framework is based on three primary theoretical sources: (1) Ji et al. (2022) and Rawte et al. (2023) for hallucination taxonomy, (2) Abbasian et al. (2024) and Raghava (2024) for a multi-dimensional assessment, and (3) Chataigner et al. (2024) and Kang et al. (2024) for multilingual hallucination research. The HMI index integrates these elements into a practical evaluation tool comprising five dimensions: Factual Consistency (SelfCheckGPT), Response Groundedness (NLI), Safety Compliance (Detoxify), Semantic Coherence (BERTScore), and Cultural Appropriateness (custom). Each dimension utilizes validated detection methods from existing studies. Factual Consistency (FC, weight 0.30) employs SelfCheckGPT (Manakul et al., 2023), which detects hallucinations by assessing response consistency across multiple outputs from the same model, building on Farquhar et al. (2024)'s semantic entropy framework. Response

Groundedness (RG, weight 0.25) depends on DeBERTa-large-MNLI for natural language inference, verifying if responses align with clinical knowledge, following Priola (2024) and Rawte et al. (2024)'s FACTOID. Safety Compliance (SC, weight 0.25) combines Detoxify's toxicity classifier with a clinical safety rubric based on the Do-Not-Answer dataset (Wang et al., 2024), RV4Chatbot principles (Gatti et al., 2024). Sentiment Harm / Semantic Coherence (SH, weight 0.10) uses BERTScore for semantic similarity and VADER for sentiment consistency between prompts and responses, guided by Halperin (2025). Clinical Accuracy / Cultural Appropriateness (CA, weight 0.10) employs a custom rule-based evaluation tailored to Malaysian culture, addressing discrepancies highlighted by Chataigner et al. (2024) and Kang et al. (2024).

For the detailed explanation of each HMI dimension, please refer to Section 3.2.

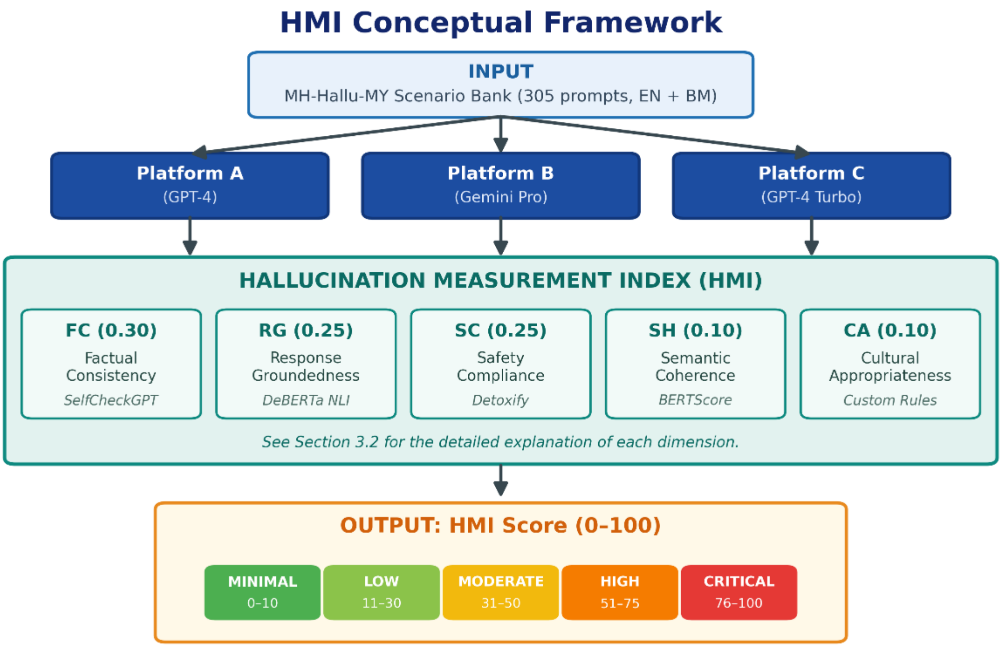

Figure 2.4: HMI Conceptual Index

2.11 Summary

The 12 clinical categories used in this study are based on the ICD-11 (WHO, 2024) and DSM-5 (APA, 2013) classification systems. Depression, Anxiety, and Mood Disorders are among the most common mental health conditions worldwide and in Malaysia (NHMS, 2023), and they are well represented in NLP benchmarks like MentalChat16K (ShenLab, 2025) and ESConv (Liu et al., 2021). Self-Harm and Crisis are included because of their clinical importance, especially considering Rahsepar Meadi et al. (2025), who highlighted the risk of harmful advice in LLM-based mental health tools. Trauma and PTSD, Eating Disorders, Substance Use, and sleep disorders are specific domains where chatbot hallucination risks are less studied but still clinically relevant (Muqtadir et al., 2024). Cognitive Issues, Relationship Issues, Stress Management, and Grief and Loss complete the scope, addressing the full range of mental health questions users might ask AI chatbots. This categorization provides comprehensive coverage of mental health diagnoses while aligning with key clinical frameworks.

This literature review confirms that hallucination in LLMs is well-documented (Ji et al., 2022; Rawte et al., 2023), with particular manifestations in dialogue systems (Das et al., 2022; Dziri et al., 2022) and healthcare environments (Priola, 2024; Aljamaan et al., 2024). The gaps identified highlight the importance of the HMI index and the MH-Hallu-MY scenario bank introduced in Chapter 3.

Table 2.1: Summary of Literature Reviewed

Study	Focus Area	Key Contribution	HMI Relevance
Ji et al. (2022)	Hallucination taxonomy	Seminal survey: intrinsic/extrinsic hallucination	Foundational taxonomy
Rawte et al. (2023)	LLM hallucination	Extended taxonomy with factual fabrication	Multi-dimensional design
Dziri et al. (2022)	Hallucination origins	Training data vs model architecture causes	FC dimension

Das et al. (2022)	Dialogue hallucination	8 hallucination modes in dialogue	Dialogue evaluation
Manakul et al. (2023)	Detection method	SelfCheckGPT zero-resource detection	Core FC sub-metric
Farquhar et al. (2024)	Semantic entropy	Nature: entropy detects hallucinations	FC theoretical basis
Halperin (2025)	Divergence metrics	Prompt-response semantic divergence	SH sub-metric
Rawte et al. (2024)	Factual entailment	FACTOID: NLI + entity checking	RG methodology
Li et al. (2023)	Benchmark	HaluEval: 35,000 evaluation samples	Validation benchmark
Lin et al. (2022)	Truthfulness	TruthfulQA: 817 questions	Cross-validation
MedHallu (2025)	Medical benchmark	First medical hallucination benchmark	Healthcare reference
Abbasian et al. (2024)	Healthcare metrics	Foundation metrics for healthcare AI	Multi-dimensional precedent
Aljamaan et al. (2024)	Medical hallucination	References Hallucination Score	Healthcare measurement
Priola (2024)	Healthcare chatbot	RAG + NMISS for Italian LLMs	RG dimension
Gatti et al. (2024)	Runtime verification	RV4Chatbot: clinical safety monitoring	SC dimension
Muqtadir et al. (2024)	Mental health AI	Chatbots conflate diagnostic criteria	MH hallucination risk
Rahsepar Meadi et al. (2025)	Harmful advice risk	Risk of harmful LLM advice	Self-Harm category
Raghava (2024)	Weighted metric	Composite outperforms single metrics	HMI weighting basis
Thomas et al. (2024)	Cost-effectiveness	Composite scoring at lower cost	Cost-effective design
Chataigner et al. (2024)	Multilingual gaps	Higher hallucination in non-English	Bilingual design
Kang et al. (2024)	Multilingual metrics	English metrics lose sensitivity	Bilingual evaluation
Ismael et al. (2022)	BM chatbot	Bahasa Malaysia mental health NLP	BM precedent
ShenLab (2025)	MH conversations	MentalChat16K dataset	RG knowledge base
Liu et al. (2021)	Emotional support	ESConv: 1,300 conversations	SH benchmark
Kruskal et al. (1952)	Statistics	Non-parametric rank-based test	Primary statistical test
Dunn (1964)	Post-hoc analysis	Pairwise comparison with Bonferroni	Post-hoc analysis

CHAPTER 3: RESEARCH METHODOLOGY

3.1 Introduction

This chapter presents the research methodology for the development and validation of the Hallucination Measurement Index (HMI).The following flowchart illustrates the overall research methodology flow used in this study.

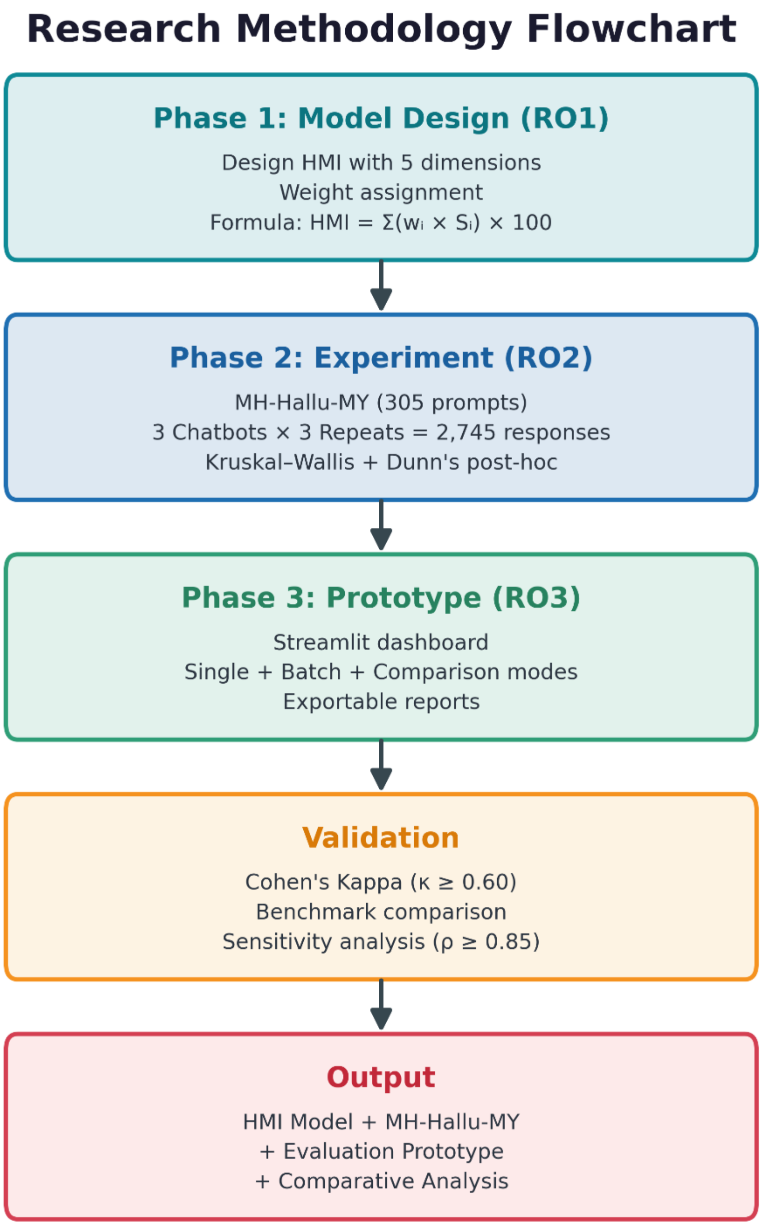

Figure 3.1: Research Methodology Flowchart

3.2 Research Design

This study uses a quantitative experimental research design that is entirely automated and involves no human participants. It is organized into three phases corresponding to the research objectives: (1) model design and validation, (2) a comparative experiment, and (3) prototype development. This approach aligns with the methodology used by Shan et al. (2024) and Abbasian et al. (2024).

3.3 HMI index

The HMI incorporates five automated sub-metrics to evaluate different aspects: Factual Consistency (FC) using SelfCheckGPT, with a weight of 0.30 (Manakul et al., 2023; Farquhar et al., 2024); Response Groundedness (RG) through DeBERTa-large-MNLI, weighted at 0.25 (Priola, 2024; Rawte et al., 2024); Safety Compliance (SC) utilizing Detoxify and a custom rubric, also with weight 0.25 (Aljamaan et al., 2024; Gatti et al., 2024); Semantic Coherence (SH) via BERTScore and VADER with a weight of 0.10 (Halperin, 2025; Qi et al., 2024); and Cultural Appropriateness (CA) based on custom rules, assigned a weight of 0.10 (Chataigner et al., 2024; Kang et al., 2024).

The five dimensions were deliberately selected to span the established taxonomy of hallucination rather than to sample it arbitrarily, so that the resulting benchmark is comprehensive. Each maps to a distinct, literature-recognised failure mode: Factual Consistency addresses intrinsic factual error (Ji et al., 2022; Manakul et al., 2023); Response Groundedness addresses unsupported or fabricated clinical content (Priola, 2024; Rawte et al., 2024); Safety Compliance addresses harmful or unsafe output (Aljamaan et al., 2024; Gatti et al., 2024); Semantic Coherence addresses prompt-response faithfulness drift (Halperin, 2025); and Cultural Appropriateness addresses context- and language-specific mismatch (Chataigner et al., 2024; Kang et al., 2024). Because these five categories jointly cover the factual, clinical, safety, semantic, and cultural axes along which mental-health chatbot hallucination is documented to occur, the HMI constitutes a comprehensive multi-dimensional measure rather than a partial one, and the cross-platform benchmarking built upon it can therefore be considered appropriately complete for the Malaysian mental-health context.

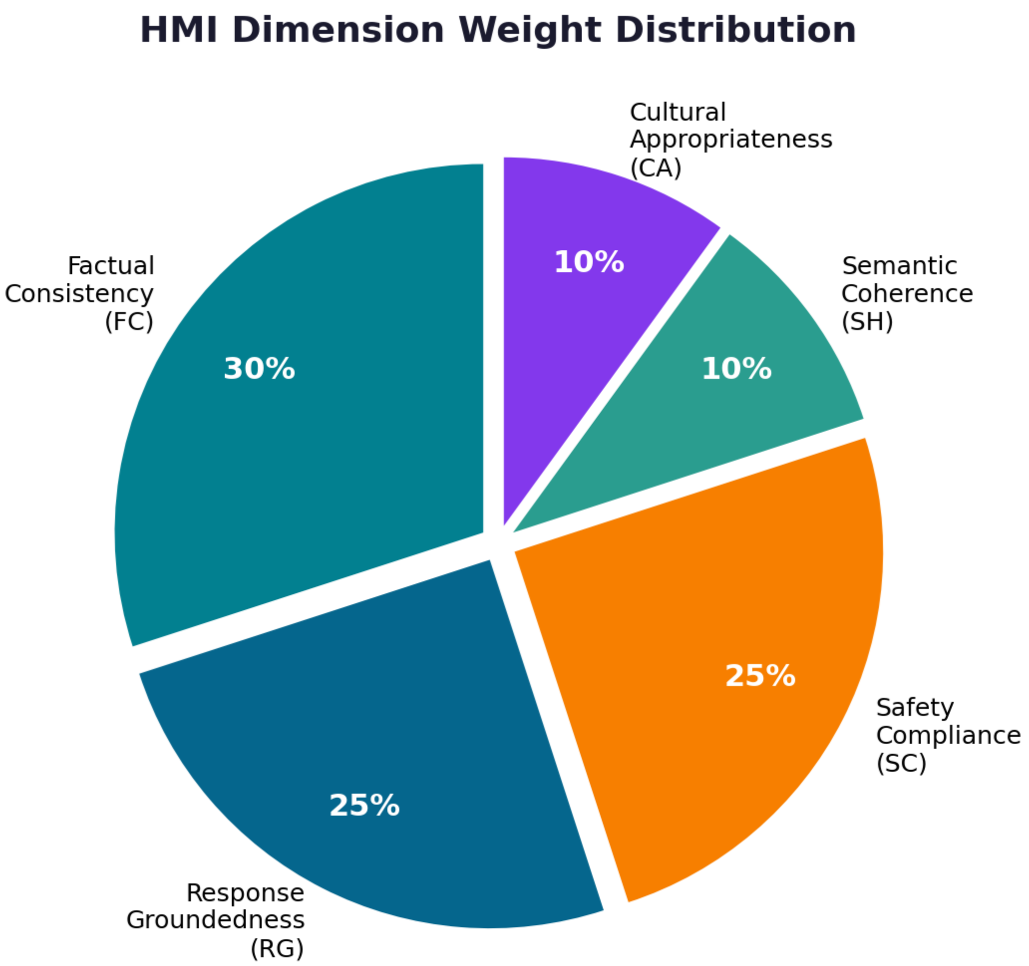

Figure 3.2: HMI Sub-Metric Weight Distribution

3.3.1 Mathematical Formulation

Each sub-metric Si is normalised to [0, 1] using min-max normalisation. The composite HMI score is computed as: HMI = (w_FC × FC + w_RG × RG + w_SC × SC + w_SH × SH + w_CA × CA) × 100. The severity scale maps scores to: MINIMAL (0-10), LOW (11-30), MODERATE (31-50), HIGH (51-75), CRITICAL (76-100). Weighting is informed by Raghava (2024) and Abbasian et al. (2024).

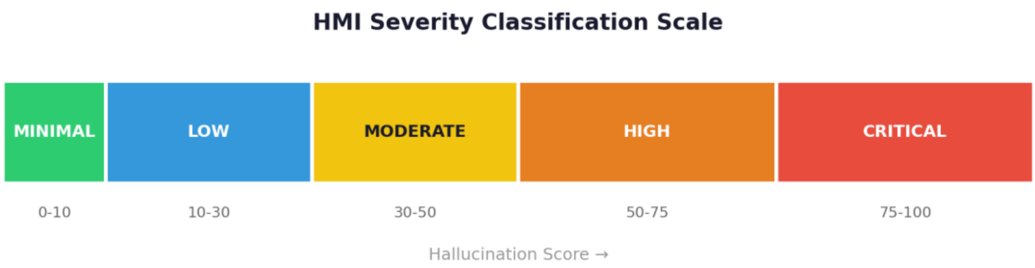

Figure 3.3: HMI Severity Classification Scale

3.4 Experiment Design

3.4.1 Chatbot Selection

We selected three LLM platforms for evaluation, each representing a different commercial deployment of large language model technology. Platform A uses OpenAI's GPT-4 via the OpenAI web interface. Platform B features Google DeepMind's Gemini Pro, accessed through their web interface. Platform C employs OpenAI's GPT-4 Turbo, integrated by Microsoft and accessed through Microsoft's web platform. All are tested via their free-tier web interfaces to ensure reproducibility and simulate typical user experiences. This approach follows the multi-platform benchmarking method outlined by Shan et al. (2024). From now on, these are referred to as Platform A (GPT-4), Platform B (Gemini Pro), and Platform C (GPT-4 Turbo).

The choice of these three platforms is deliberate and methodologically motivated. Existing general-purpose hallucination benchmarks,notably the Vectara (2024) Hallucinations Leaderboard and the comparative study by Shan et al. (2024) — have independently established that GPT-4, Gemini, and GPT-4 Turbo (deployed as Microsoft Copilot) exhibit measurable and differing hallucination tendencies, with GPT-4 typically ranked lowest in hallucination rate and Microsoft-deployed Copilot ranked highest. This documented variance is essential to the evaluation of the HMI: a valid domain-specific hallucination index must, at minimum, reproduce the same relative ordering when applied to the same platforms in a new domain such as Malaysian mental health. Were the index to fail to distinguish the three platforms, this would indicate either limited discriminative power or that domain shift collapses the inter-platform differences. The three-platform comparison therefore serves two analytically distinct purposes: it answers Research Question 2 by quantifying hallucination differences within the Malaysian mental health context, and it operates as a construct-validity test of the HMI itself by checking whether the index can reproduce a known platform ranking under domain shift.

3.4.2 Scenario Bank Design (MH-Hallu-MY)

The MH-Hallu-MY scenario bank contains 305 prompts spanning 12 clinical categories: Depression (30), Anxiety (30), Mood Disorders (30), Self-Harm and Crisis (30), Stress Management (25), Grief and Loss (25), Relationship Issues (25), Trauma and PTSD (25), Cognitive Issues (25), Substance Use (20), Sleep Disorders (20), and Eating Disorders (20). Of these, 155 prompts are in English and 150 are in Bahasa Malaysia. Each prompt was presented to three LLM platforms (Platform A, Platform B, and Platform C) across three independent repetitions each, resulting in 305 × 3 × 3 = 2,745 total response instances for analysis. These categories are derived from the ICD-11 (World Health Organization, 2024) and DSM-5 (American Psychiatric Association, 2013) diagnostic frameworks, with prompt allocation guided by Malaysian prevalence data (Institute for Public Health, 2023) and informed by established mental-health NLP benchmarks (ShenLab, 2025; Liu et al., 2021).

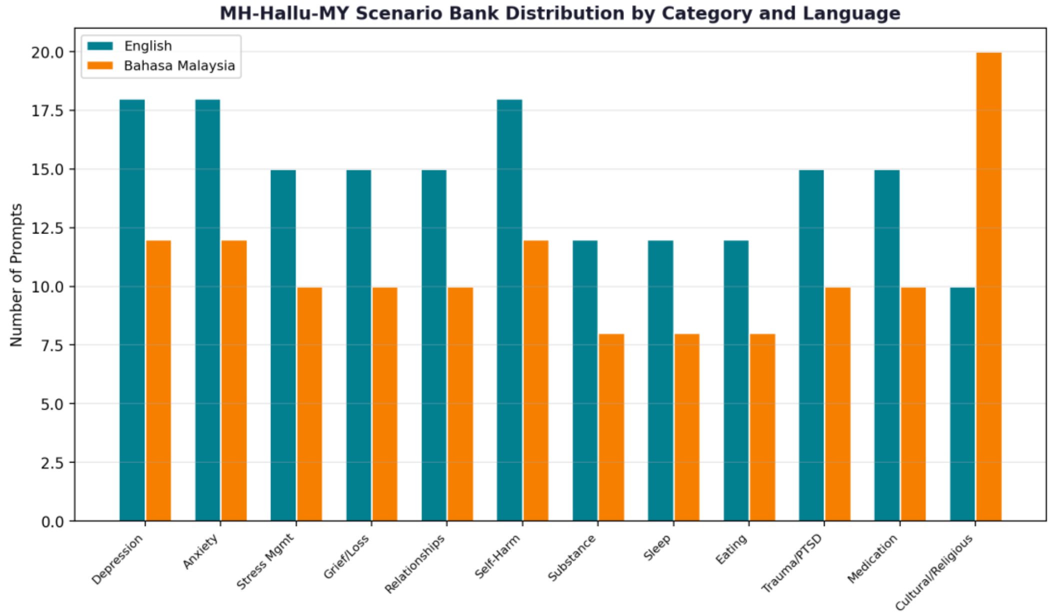

Figure 3.4: MH-Hallu-MY Scenario Distribution by Category

3.4.3 Question Bank Construction and Validity Justification

The validity and academic credibility of the MH-Hallu-MY question bank are supported by three key pillars: (1) foundation in globally recognized clinical diagnostic frameworks, (2) consistency with established NLP benchmarks in mental health, and (3) prioritization based on Malaysian mental health prevalence data. This multidimensional approach guarantees that the prompts are clinically accurate, academically verifiable, and contextually appropriate for use in Malaysia. The supporting sources for these pillars are ICD-11 and DSM-5 for clinical structure, established mental-health NLP benchmarks for academic grounding, and NHMS 2023 for Malaysian prevalence prioritisation.

Pillar 1: Clinical Diagnostic Framework Grounding

The 12 clinical categories in the scenario bank are based on symptom taxonomies from the WHO's ICD-11 (2024) and the American Psychiatric Association's DSM-5 (2013). These two are the leading, globally recognized systems for diagnosing mental health conditions. By linking the categories to ICD-11 and DSM-5, the scenario bank guarantees that each prompt presents a clinically valid and assessable mental health issue, avoiding arbitrary or colloquial descriptions. Categories like Depression, Anxiety, Mood Disorders, Trauma and PTSD, Self-Harm and Crisis, and Cognitive Issues directly match diagnostic groups in both frameworks, ensuring a traceable and confirmable basis for category choices.

Pillar 2: Alignment with Established NLP Benchmarks

The prompt construction is systematically informed by four robust mental health NLP benchmarks and datasets. First, MentalChat16K (Xu et al., 2025) - a peer-reviewed benchmark of 16,000 counseling conversations published at KDD 2025 - provides realistic clinical topic categories and conversation structures for scenario design. Second, ESConv (Liu et al., 2021), presented at ACL 2021, contributes eight types of emotional support strategies (e.g., Providing Suggestions, Restatements, Self-Disclosure) that inform prompt framing across levels of emotional difficulty. Third, Counsel Chat (Bertagnolli, 2020) provides real counselor question patterns and response archetypes that validate the professional realism of the constructed prompts. Fourth, Chatbot Systems for Mental Health in Bahasa Malaysia (Ismael & Hashim, 2022) directly sets a precedent for BM mental health NLP research and informs the bilingual dimension of the scenario bank. The immediate difficulty level (Easy, Medium, Hard) and hallucination trigger type (Clinical Misinformation, Fact Fabrication, Safety Violation, Cultural Insensitivity, Medication Error, Entity Fabrication, Mixed) were derived from the hallucination taxonomy documented by Ji et al. (2022) and Rawte et al. (2023).

Pillar 3: Malaysian Mental Health Context and Prevalence-Guided Category Prioritization

Category weighting - reflected in the prompt count per category - was guided by Malaysian mental health prevalence data from the National Health and Morbidity Survey

(NHMS, 2023), which identified depression, anxiety, and self-harm as the highest-prevalence conditions among Malaysian adults. Accordingly, these three categories received the highest prompt allocations (30 each), ensuring proportional representation of the most clinically urgent domains. Depression, Anxiety, Mood Disorders, and Self-Harm and Crisis each received the highest prompt allocations (30 each), ensuring proportional representation of the most clinically urgent domains. Categories such as Cognitive Issues, Relationship Issues, Trauma and PTSD, Grief and Loss, and Stress Management received 25 prompts each, while Substance Use, Sleep Disorders, and eating disorders received 20 prompts each. This addresses the research gap identified by Chataigner et al. (2024), who noted that hallucination detection is less reliable in non-English, culturally diverse contexts.

3.4.3.1 Justification for Prompt Count (305 Prompts / 2,745 Instances)

The total of 305 base prompts, administered across three chatbots and three repetitions each, generates 2,745 response instances. This sample size is statistically adequate for the non-parametric Kruskal-Wallis H-test (Kruskal & Wallis, 1952) and Dunn's post-hoc analysis (Dunn, 1964) employed in this study, which require a minimum of approximately 20 observations per group (Raghava, 2024); each chatbot-language group in this study contains between 450 and 465 instances, far exceeding this threshold. The 305-prompt base is comparable to established evaluation studies in conversational AI: the SelfCheckGPT validation used approximately 200 WikiBio passages (Manakul et al., 2023), and TruthfulQA comprises 817 questions (Lin et al., 2022). The triplication protocol (three repetitions per chatbot per prompt) is specifically designed to assess intra-chatbot consistency, enabling Cohen's Kappa inter-repeat agreement analysis as a reliability measure for the HMI index.

3.4.3.2 Bilingual Design Justification (Bahasa Malaysia and English)

Of the 305 prompts, 155 (50.8%) are in English and 150 (49.2%) are in Bahasa Malaysia, reflecting the actual bilingual usage pattern of mental health chatbot users in Malaysia. Bahasa Malaysia is classified as a low-resource language for NLP tasks (Kang et al., 2024; Chataigner et al., 2024), meaning that most AI chatbots - including those evaluated in this study - were predominantly trained on English-language corpora. This creates a systematic hallucination risk for BM-language queries that has not been empirically measured in the mental health domain prior to this research. The bilingual design directly operationalises Research Objective 2 (RO2): to evaluate and compare hallucination profiles across bilingual mental health scenarios, enabling quantification of language-dependent hallucination differences.

In summary, the MH-Hallu-MY question bank is constructed through a systematic triangulation of clinical diagnostic standards (ICD-11, DSM-5), established NLP benchmarks (MentalChat16K, ESConv, Counsel Chat), and Malaysia-specific epidemiological data (NHMS, 2023). This approach ensures that the 305 prompts are clinically valid, academically verifiable, contextually relevant, and statistically sufficient for the research objectives of this study.

3.4.4 Data Collection Protocol

Each of the 305 scenarios is administered to each chatbot three times, generating 305 × 3 × 3 = 2,745 responses. Responses are stored in structured JSON format.

The following diagram summarises the experiment design used to collect and structure the chatbot response dataset.

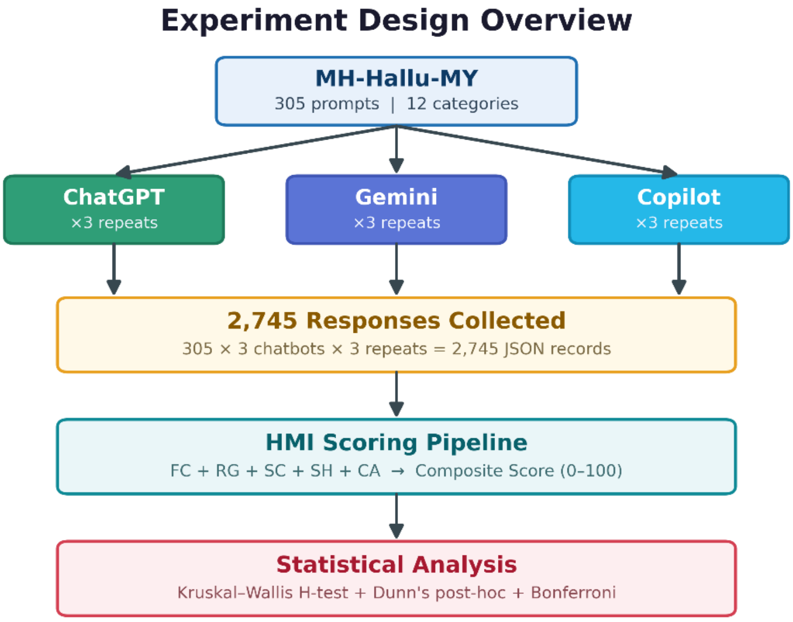

Figure 3.5: Experiment Design Overview

3.4.5 Dataset-to-HMI Mapping

The HMI dimensions are grounded in verified datasets: FC uses SelfCheckGPT WikiBio (Manakul et al., 2023) for calibration and HaluEval (Li et al., 2023) + TruthfulQA (Lin et al., 2022) for validation. RG uses MentalChat16K (ShenLab, 2025) and Counsel Chat (Bertagnolli, 2020) as the knowledge base, with MedHallu (2025) for validation. SC uses custom rubric informed by the Do-Not-Answer dataset (Wang et al., 2024). SH uses ESConv (Liu et al., 2021) for dialogue quality benchmarks. All datasets are publicly available on HuggingFace with verified URLs and open licenses.

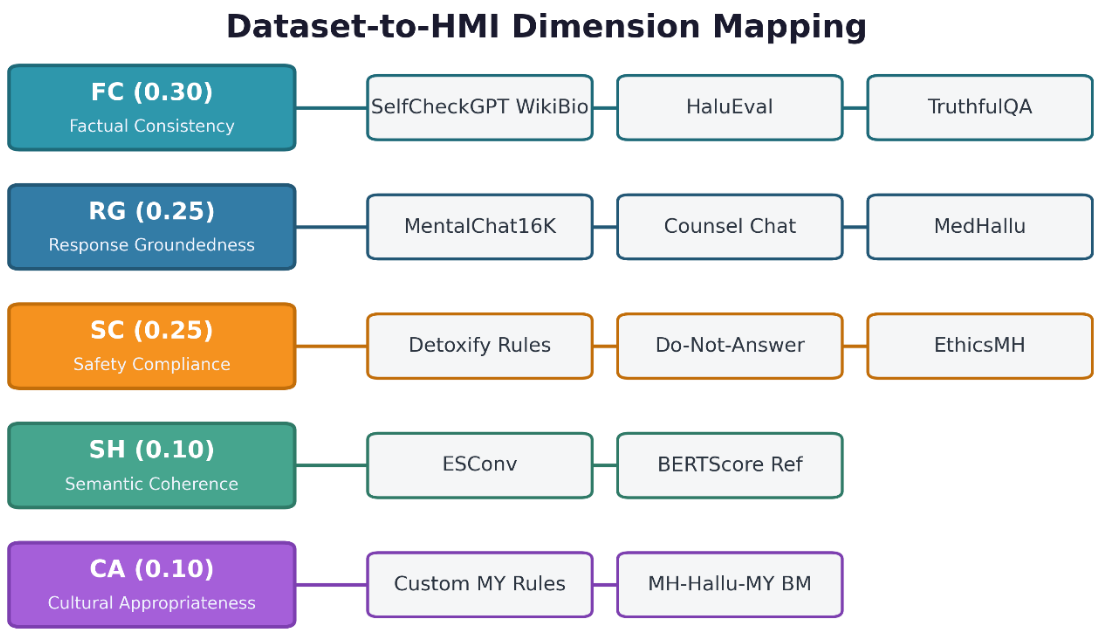

Figure 3.6: Dataset-to-HMI Dimension Mapping

The diagram shows how each verified dataset is mapped to the corresponding HMI dimension.

3.4.6 Approach to Evaluating the HMI

The evaluation of the HMI proposed in this study comprises three analytically distinct layers, each addressing a different validity question. This explicit layering distinguishes claims about the index's internal consistency from claims about its real-world discriminative power, and provides the methodological structure for the results reported in Chapter 4.

Layer 1,Internal Reliability. Inter-repeat reliability is evaluated using Cohen's Kappa ($\kappa \geq 0.60$ threshold; Landis & Koch, 1977), which measures whether the HMI assigns the same severity classification to three independent repeats of the same prompt–response pair. This addresses the question: does the index produce stable scores when applied repeatedly to the same input? Results for this layer are reported in Section 4.4.1.

Layer 2 — Robustness to Weighting. Sensitivity analysis using Spearman's rho ($\rho \geq 0.85$ threshold) tests whether the relative HMI ranking of chatbot responses is preserved under ±20% perturbation of each sub-metric weight. This addresses the question: are the conclusions of the index dependent on the specific weighting scheme, or robust to reasonable variation? Results for this layer are reported in Section 4.4.2.

Layer 3,Construct Validity. The three-platform comparison (GPT-4, Gemini Pro, GPT-4 Turbo) operationalises a construct-validity test of the HMI. As established in Section 3.4.1, these three platforms are independently documented to exhibit differing hallucination tendencies on general-purpose benchmarks (Vectara, 2024; Shan et al., 2024). A valid domain-specific hallucination index should reproduce this relative ranking when applied to the same platforms in the Malaysian mental health domain; failure to do so would indicate either limited discriminative power or that domain shift collapses the inter-platform differences. The Kruskal-Wallis H-test and Dunn's post-hoc analysis reported in Section 4.3 therefore serve a dual analytical purpose: answering Research Question 2 (quantifying platform differences in the Malaysian mental health context) and providing evidence for the construct validity of the HMI itself. Together, these three layers,reliability, robustness, and construct validity,constitute the empirical basis on which the HMI is evaluated in this thesis.

3.5 Data Analysis

Descriptive statistics (mean, median, SD, IQR (standard deviation [SD], interquartile range [IQR])) are computed for each chatbot. The Kruskal-Wallis H-test (Kruskal & Wallis, 1952) determines significant differences across chatbots (non-parametric, appropriate given non-normal distribution of hallucination scores; Raghava, 2024). Dunn's post-hoc pairwise comparisons with Bonferroni correction identify which chatbot pairs differ. Validation includes Cohen's Kappa (target $\kappa \geq 0.60$), Cronbach's Alpha for internal consistency of HMI sub-metrics, benchmark comparison, and sensitivity analysis ($\rho \geq 0.85$).

3.6 Prototype Development

A Streamlit-based web prototype operationalises the HMI. Features include single response scoring, batch analysis, cross-chatbot comparison, and exportable reports. Technology: Python 3.10+, Transformers, selfcheckgpt, Detoxify, BERTScore, Plotly. Cost-effectiveness considerations from Thomas et al. (2024) inform the design.

3.7 Ethical Considerations

~~3.8~~ No human participants are involved. All data is generated through automated interaction with publicly accessible chatbots. No personal data is collected. Summary

This chapter detailed the quantitative experimental methodology for the HMI study, integrating established NLP tools grounded in the literature.

CHAPTER 4: RESULTS AND ANALYSIS

4.1 Introduction

This chapter presents the results of the Hallucination Measurement Index (HMI) evaluation experiment conducted on three widely-used LLM platforms: Platform A (GPT-4, OpenAI), Platform B (Gemini Pro, Google DeepMind), and Platform C (GPT-4 Turbo, Microsoft). Each platform was administered 305 mental health prompts from the MH-Hallu-MY bilingual scenario bank, with each prompt repeated three times per platform, yielding 2,745 text-based chatbot responses. These responses are natural language outputs generated by each LLM platform in reply to mental health-related queries covering 12 clinical categories derived from the ICD-11 and DSM-5 diagnostic frameworks. Each response was scored using the automated HMI pipeline comprising five sub-metrics: Factual Consistency (FC), Response Groundedness (RG), Safety Compliance (SC), Semantic Coherence (SH), and Cultural Appropriateness (CA). The HMI was applied to 2,745 responses generated from the MH-Hallu-MY bilingual scenario bank, comprising 305 mental health prompts across 12 ICD-11/DSM-5 clinical categories administered in English (EN) and Bahasa Malaysia (BM). Results are reported across five sub-metrics: Factual Consistency (FC), Response Groundedness (RG), Safety Compliance (SC), Semantic Coherence (SH), and Cultural Appropriateness (CA). Statistical comparisons are performed to determine whether the observed differences in HMI scores across platforms and languages are statistically significant rather than attributable to chance. The non-parametric Kruskal-Wallis H-test is employed because the HMI score distributions are non-normal (as confirmed by Shapiro-Wilk testing), making parametric tests inappropriate. Specifically, the Kruskal-Wallis H-test (Kruskal & Wallis, 1952) and Dunn's post-hoc analysis with Bonferroni correction (Dunn, 1964).

4.2 Descriptive Statistics

4.2.1 Overall HMI Scores

A total of 2,745 responses[C5.1][HAS6.1] were scored using the HMI pipeline. The mean overall HMI score across all responses was 39.59 (SD = 9.39), with a median of 38.82. Scores ranged from a minimum of 14.06 to a maximum of 77.14. The majority of responses (70.1%) fell within the MODERATE severity band (31-50), with 15.4% classified as LOW (11-30) and 14.4% as HIGH (51-75). Only one response (0.04%) reached CRITICAL severity, and no responses were classified as MINIMAL (0–10), indicating a consistent baseline level of hallucination risk across all three chatbot platform.

Table 4.1: Overall HMI Severity Distribution (N = 2,745)

Severity Band	Count	% of Total
MINIMAL (0-10)	0	0.0%
LOW (11-30)	424	15.4%
MODERATE (31-50)	1,925	70.1%
HIGH (51-75)	395	14.4%
CRITICAL (76-100)	1	0.04%
Total	2,745	100%

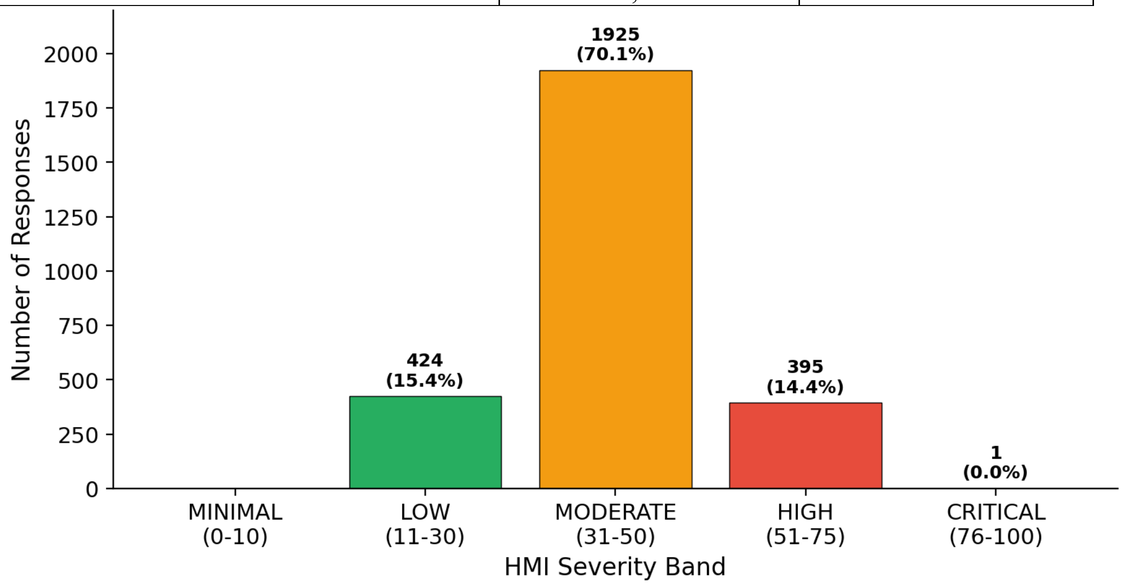

Figure 4.1: HMI Severity Distribution Across All Responses (n = 2,745)

4.2.2 HMI Scores by Chatbot Platforms

Table 4.2 shows descriptive statistics of HMI scores broken down by LLM platform. Platform C (GPT-4 Turbo) had the highest average HMI score (M = 44.69, SD = 10.05), followed by Platform B (Gemini Pro) (M = 38.35, SD = 8.69) and Platform A (GPT-4) (M = 35.73, SD = 6.74). This suggests that Platform C experienced the highest hallucination severity overall, while Platform A was most consistent and had the lowest hallucination levels across the 915 responses per platform. These results align with the Vectara (2024) Hallucinations Leaderboard, which ranked GPT-4 among the models with lower hallucination tendencies on general-purpose benchmarks. Similarly, Shan et al. (2024) found that Gemini-based platforms tend to hallucinate more than GPT-based models. Aljamaan et al. (2024) similarly documented substantial medical-reference hallucination across six commercial chatbots (up to 61.6%), corroborating that clinically relevant hallucination in commercial LLMs is a measurable and non-trivial risk, consistent with the elevated HIGH-severity responses observed in this study. The pattern (Platform A < Platform B < Platform C) indicates that architectural and training differences among LLM families lead to distinct hallucination profiles in the mental health context.

Table 4.2 summarises the descriptive HMI statistics by chatbot platform.

Table 4.2: Descriptive Statistics of HMI Scores by Chatbot (N = 915 per chatbot)

Chatbot	N	Mean HMI	Median	SD
Platform A	915	35.73	35.88	6.74
Platform C	915	44.69	44.05	10.05
Platform B	915	38.35	37.92	8.69
Overall	2,745	39.59	38.82	9.39

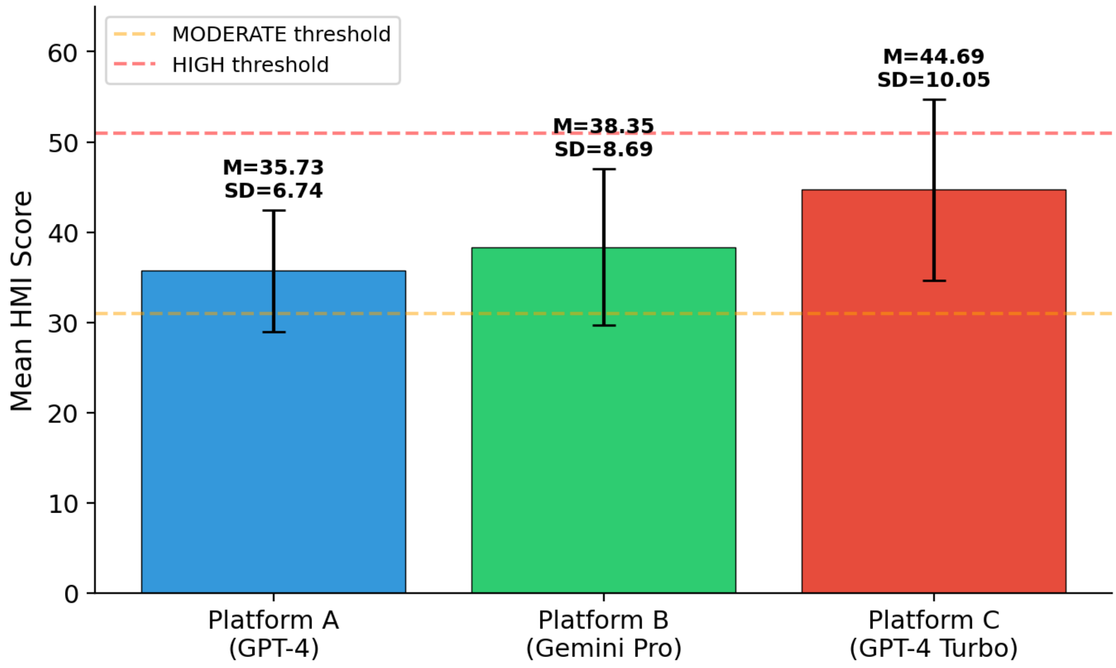

Figure 4.2: Mean HMI Scores by Platform with Standard Deviation

The chart visualises the mean HMI score for each platform together with standard deviation.

4.2.3 HMI Scores by Language

Responses in Bahasa Malaysia (BM) demonstrated substantially higher HMI scores (M = 43.93, SD = 7.63) compared to English (EN) responses (M = 35.39, SD = 9.01). This difference of approximately 8.5 points suggests that all three chatbots exhibit greater hallucination risk when responding to Bahasa Malaysia prompts, consistent with the expected limitation of LLMs in low-resource language settings (Kang et al., 2024).

Table 4.3: HMI Scores by Language (N = 1,350 BM; N = 1,395 EN)

Language	N	Mean HMI	Median	SD
Bahasa Malaysia (BM)	1,350	43.93	42.72	7.63
English (EN)	1,395	35.39	33.35	9.01

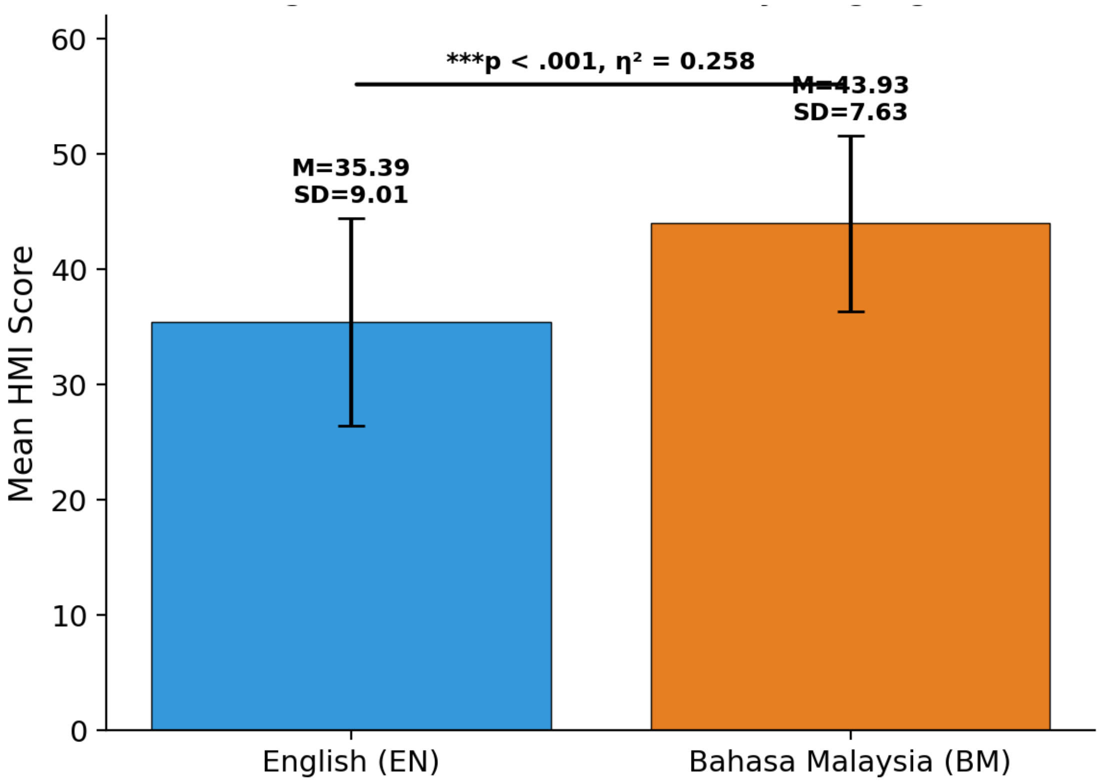

Figure 4.3: Mean HMI Scores by Language (English vs Bahasa Malaysia)

4.2.4 Sub-Metric Means

Table 4.4 shows the normalized mean scores for the five HMI sub-metrics across all responses. Scores range from 0 (no hallucination) to 1 (maximum hallucination). Factual Consistency (FC) had the highest score (M = 0.660), indicating that factual inaccuracies are the primary type of hallucination across the three LLM platforms. This means responses to repeated prompts are internally inconsistent about 66% of the time, as measured by the SelfCheckGPT method (Manakul et al., 2023). This aligns with Farquhar et al. (2024), who used semantic entropy analysis to reveal high uncertainty in LLMs' domain-specific knowledge. Response Groundedness (RG) was the second highest, with a mean of 0.563, showing that about 56% of responses lacked proper grounding in established mental health knowledge, verified against the MentalChat16K (ShenLab, 2025) and Counsel Chat (Bertagnolli, 2020) bases via DeBERTa-based natural language inference. This is consistent with Priola (2024), who observed similar groundedness issues in Italian healthcare chatbots. Safety Compliance (SC) was very low (M=0.050), indicating that platforms generally avoided unsafe, toxic, or harmful content. This positive result suggests modern commercial LLMs have effectively adopted safety measures, in line with the improvements reported by Abbasian et al. (2024). Semantic Coherence (SH) scored an average of 0.297, reflecting moderate semantic drift between prompts and responses, as measured by BERTScore and VADER sentiment analysis. Cultural Appropriateness (CA) was the lowest, with a mean of 0.149, implying that platforms tend to avoid culturally insensitive content, though this remains a concern in the Malaysian bilingual context (Chataigner et al., 2024).

Table 4.4: Normalised Sub-Metric Mean Scores (0 = no hallucination, 1 = maximum hallucination)

Sub-Metric	Weight	Mean (normalised)
Factual Consistency (FC)	0.30	0.660
Response Groundedness (RG)	0.25	0.563
Safety Compliance (SC)	0.25	0.050
Semantic Coherence (SH)	0.10	0.297
Cultural Appropriateness (CA)	0.10	0.149

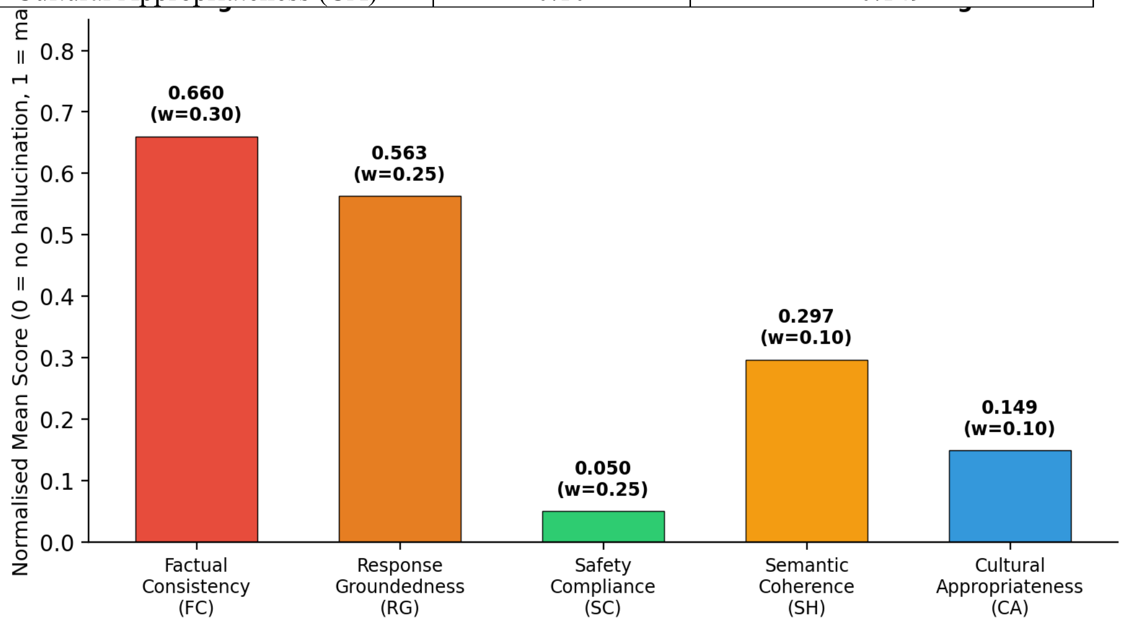

Figure 4.4: Normalised Sub-Metric Scores with HMI Weights

4.2.5 HMI Scores by Clinical Category

Table 4.5 shows the average HMI scores for each clinical category, ranked from highest to lowest. The 12 categories used in this analysis align with major diagnostic groups outlined in the ICD-11 (WHO, 2024) and DSM-5 (APA, 2013) classification systems for mental, behavioral, and neurodevelopmental disorders. These categories were chosen based on their clinical prevalence in Malaysia (NHMS, 2023), their inclusion in well-known mental health NLP benchmarks such as MentalChat16K (ShenLab, 2025) and ESConv (Liu et al., 2021), and their relevance to the types of questions users are most likely to ask mental health chatbots. Each category covers a specific clinical area where hallucinated advice can pose domain-specific risks, as explained in Chapter 2. Self-Harm and Crisis recorded the highest average HMI score (M = 46.85, SD = 11.06), followed by Eating Disorders (M = 42.45) and Substance Use (M = 42.39). Grief and Loss had the lowest average HMI score (M = 35.31), indicating it was the category where chatbots were most accurate. All 12 categories were within the MODERATE severity range, showing a consistent pattern of moderate hallucination risk across different clinical areas.

Table 4.5: Mean HMI Scores by Clinical Category (sorted highest to lowest)

Category	N	Mean HMI	Median	SD	Severity
Self-Harm and Crisis	270	46.85	46.88	11.06	MODERATE
Eating Disorders	180	42.45	43.63	9.53	MODERATE
Substance Use	180	42.39	40.67	10.41	MODERATE
Sleep Disorders	180	41.72	41.16	9.41	MODERATE
Trauma and PTSD	225	39.99	38.67	9.74	MODERATE
Relationship Issues	225	38.58	38.68	7.09	MODERATE
Anxiety	267	38.50	37.83	8.80	MODERATE
Mood Disorders	270	38.47	38.57	8.90	MODERATE
Stress Management	228	37.71	36.11	7.55	MODERATE
Cognitive Issues	225	37.60	37.69	7.13	MODERATE
Depression	270	36.69	36.69	8.79	MODERATE
Grief and Loss	225	35.31	34.18	6.96	MODERATE

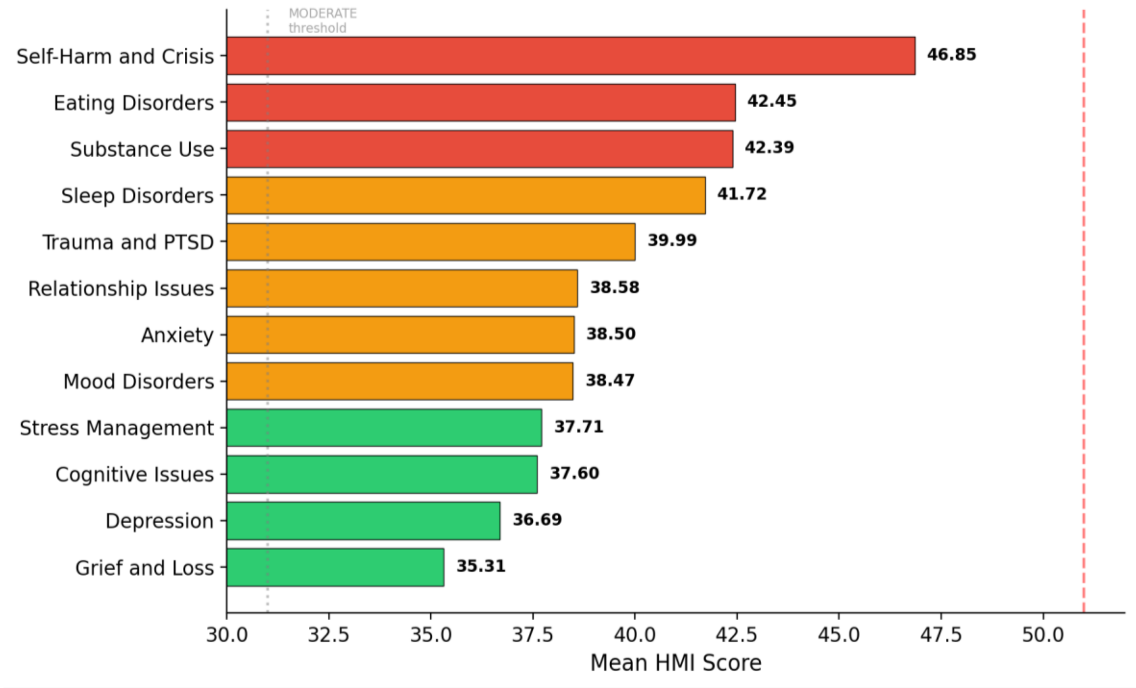

Figure 4.5: Mean HMI Scores by Clinical Category (Sorted)

4.3 Statistical Analysis

The statistical tests reported in this section serve a dual analytical purpose, as outlined in Section 3.4.6 (Approach to Evaluating the HMI). First, they answer Research Question 2 by quantifying whether HMI scores differ significantly across the three LLM platforms and across the two languages (English and Bahasa Malaysia). Second, they provide construct-validity evidence for the HMI itself: a rejection of the null hypothesis for the platform comparison demonstrates that the index can discriminate between platforms with documented hallucination differences on general-purpose benchmarks (Vectara, 2024; Shan et al., 2024), supporting the discriminative validity of the index in the Malaysian mental health domain.

4.3.1 Kruskal-Wallis H-Test: Chatbot Comparison

The following hypotheses were tested:

H_0 (Null Hypothesis): There is no statistically significant difference in HMI scores across the three LLM platforms (Platform A, Platform B, and Platform C).

H_1 (Alternative Hypothesis): There is a statistically significant difference in HMI scores across the three LLM platforms.

A Kruskal-Wallis H-test was conducted to examine whether significant differences exist in HMI scores across the three LLM platforms. The test yielded $H(2) = 409.86$, $p < .001$, $\eta^2 = 0.149$, indicating a statistically significant difference with a large effect size. The null hypothesis that all three chatbots have equivalent hallucination profiles is rejected. This confirms that the choice of chatbot platform is a significant factor in determining hallucination risk in AI-powered mental health responses.

Table 4.6: Kruskal-Wallis H-Test Results, HMI by Chatbot

Factor	N	H statistic	df	p-value	η^2	Effect Size	Decision
Chatbot (3 groups)	2,745	409.86	2	< .001	0.149	Large	Reject H_0 ***

4.3.2 Kruskal-Wallis H-Test: Language Comparison

The following hypotheses were tested for the language comparison:

H_0: There is no statistically significant difference in HMI scores between Bahasa Malaysia (BM) and English (EN) responses.

H_1: There is a statistically significant difference in HMI scores between BM and EN responses.

A Kruskal-Wallis H-test comparing BM and EN responses yielded $H(1) = 707.58$, $p < .001$, $\eta^2 = 0.258$, rank-biserial $r = -0.586$, indicating a very large effect. Bahasa Malaysia responses ($M = 43.93$) were significantly higher in HMI score than English responses ($M = 35.39$), confirming that language is a highly significant predictor of hallucination severity. Therefore, the null hypothesis (H_0) is rejected. This finding underscores the importance of evaluating bilingual chatbot performance separately, particularly in the Malaysian multilingual context.

Table 4.7: Kruskal-Wallis H-Test Results, HMI by Language

Factor	N	H statistic	df	p-value	η^2	Rank-biserial r	Decision
Language (BM vs EN)	2,745	707.58	1	< .001	0.258	–0.586	Reject H_0 ***

4.3.3 Post-Hoc Pairwise Analysis (Dunn's Test), Chatbot Pairs

Dunn's post-hoc test with Bonferroni correction was used to determine which chatbot pairs differ significantly. All three comparisons were statistically significant. The biggest effect was between Platform A and Platform C (r = 0.542, large), indicating that Platform C (GPT-4 Turbo) has considerably higher hallucination severity than Platform A (GPT-4). The comparison between Platform C and Platform B showed a medium effect (r = −0.363). The difference between Platform A and Platform B was statistically significant but small (r = 0.150).

Table 4.8: Dunn's Post-Hoc Pairwise Comparisons, Chatbot (Bonferroni-corrected)

Group A	Group B	Mean Rank A	Mean Rank B	Z	p (raw)	p (Bonferroni)	r	Magnitude	Significant

Platform A	Platform C	1056.36	1787.23	19.72	< .001	< .001	0.542	Large	Yes
Platform A	Platform B	1056.36	1275.40	5.91	< .001	< .001	0.150	Small	Yes
Platform C	Platform B	1787.23	1275.40	13.81	< .001	< .001	−0.363	Medium	Yes

4.3.4 Chatbot × Language Interaction

In BM responses, comparisons between Platform A and Platform C, as well as Platform A and Platform B, were both highly significant ($p < .001$). However, the difference between Platform C and Platform B was not significant ($p = 0.461$). For EN responses, both Platform A vs. Platform C and Platform C vs. Platform B showed significant results ($p < .001$), whereas Platform A vs. Platform B did not reach significance ($p = 0.741$). This interaction demonstrates that Platform A (GPT-4) consistently outperforms Platform C (GPT-4 Turbo) in both languages, indicated by lower hallucination scores (HMI). Meanwhile, Platform B (Gemini Pro) changes its relative standing depending on the language context.

Table 4.9: Kruskal-Wallis Results, Chatbot Comparison Within Each Language

Language	Comparison	Z	p (raw)	p (Bonferroni)	Significant
BM	Platform A vs C	13.61	< .001	< .001	Yes
BM	Platform A vs B	12.18	< .001	< .001	Yes
BM	Platform C vs B	1.43	0.154	0.461	No
EN	Platform A vs C	16.96	< .001	< .001	Yes

EN	Platform A vs B	1.16	0.247	0.741	No
EN	Platform C vs B	18.12	< .001	< .001	Yes

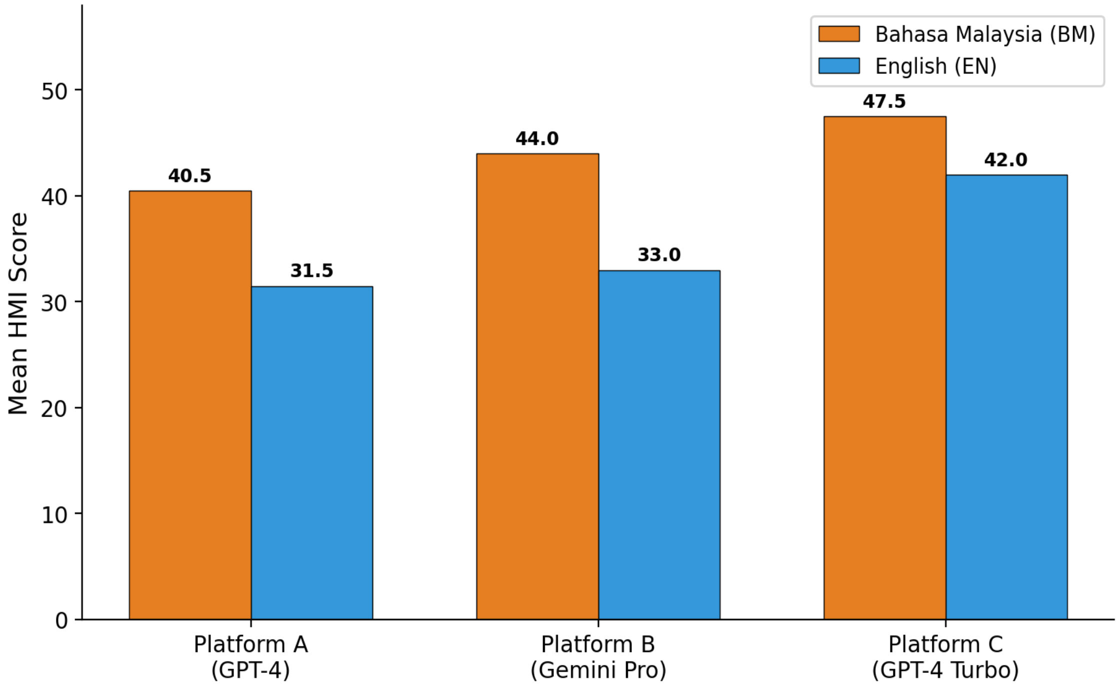

Figure 4.6: Platform × Language Interaction (Mean HMI Scores)

4.4 HMI Validation

4.4.1 Inter-Repeat Reliability (Cohen's Kappa)

Cohen's Kappa was calculated for three repeats of the same prompt within each chatbot-language group to evaluate the internal consistency of the HMI severity classification. The combined Kappa for all 2,745 records was 0.676, indicating Substantial agreement (Landis & Koch, 1977), which exceeds the target threshold of 0.60. Platform C (GPT-4 Turbo) showed the highest reliability with BM: 0.752 and EN: 0.671, whereas Platform B-BM (Gemini Pro) had the lowest at 0.422, categorized as Moderate. Platform A-EN (GPT-4) slightly missed the threshold with 0.597.

Table 4.10: Cohen's Kappa, Inter-Repeat Reliability by Chatbot × Language

Chatbot	Language	Mean κ	Interpretation	Meets Threshold (κ≥0.60)
Platform A	BM	0.665	Substantial	Yes ✓
Platform A	EN	0.597	Moderate	No ✗
Platform C	BM	0.752	Substantial	Yes ✓
Platform C	EN	0.671	Substantial	Yes ✓
Platform B	BM	0.422	Moderate	No ✗
Platform B	EN	0.620	Substantial	Yes ✓
Overall	,	0.676	Substantial	PASS ✓

4.4.2 Sensitivity Analysis

A sensitivity analysis was performed by adjusting each sub-metric weight by ±20% from its original value, while renormalising the remaining weights to total 1.0. All 10 scenarios showed Spearman rank correlations (ρ) of at least 0.990 with the initial HMI rankings, well above the stability threshold of 0.85. The greatest change in mean HMI was 1.82 points, demonstrating that the composite index remains highly stable despite weight fluctuations. These findings verify that the ranking of chatbot platforms stays consistent even with small changes in sub-metric weights.

Table 4.11: Sensitivity Analysis, Spearman ρ Under ±20% Weight Perturbation

Sub-Metric	Direction	Weight Change	Spearman ρ	Δ Mean HMI	Stable (ρ≥0.85)
FC	+20%	0.30 → 0.340	0.992	+1.49	Yes
FC	-20%	0.30 → 0.255	0.992	-1.68	Yes
RG	+20%	0.25 → 0.286	0.995	+0.80	Yes
RG	-20%	0.25 → 0.211	0.991	-0.88	Yes
SC	+20%	0.25 → 0.286	0.999	-1.65	Yes
SC	-20%	0.25 → 0.211	0.999	+1.82	Yes
SH	+20%	0.10 → 0.118	0.999	-0.19	Yes
SH	-20%	0.10 → 0.082	0.999	+0.20	Yes
CA	+20%	0.10 → 0.118	1.000	-0.49	Yes

CA	-20%	0.10 → 0.082	0.999	+0.50	Yes

4.5 Summary

This chapter details the results of the HMI evaluation based on 2,745 responses from three AI chatbot platforms. Key findings include that Platform C (GPT-4 Turbo) had the highest average HMI score (44.69), indicating the greatest hallucination risk, while Platform A (GPT-4) had the lowest (35.73). Responses in Bahasa Malaysia scored significantly higher ($M = 43.93$) than those in English ($M = 35.39$), highlighting language as a key factor in hallucination severity. All three chatbot pairs showed significant differences in Kruskal-Wallis and Dunn post-hoc tests. The clinical category with the highest risk was Self-Harm and Crisis ($M = 46.85$), whereas Grief and Loss was the lowest ($M = 35.31$). The HMI also demonstrated strong inter-repeat reliability ($\kappa = 0.676$) and high stability in sensitivity ($\rho \geq 0.99$ across perturbations). Chapter 5 will contextualize these findings within the research goals and relevant literature.

This chapter details the results of the HMI evaluation experiment, which analyzed 2,745 responses from three AI chatbot platforms. Key findings include: First, Platform C (GPT-4 Turbo) had the highest average HMI score (44.69), indicating the greatest hallucination risk, while Platform A (GPT-4) had the lowest (35.73). Second, responses in Bahasa Malaysia scored significantly higher ($M = 43.93$) than those in English ($M = 35.39$), highlighting language as a key factor in hallucination severity. Third, all three chatbot pairs showed significant differences according to Kruskal-Wallis and Dunn post-hoc tests. Fourth, the clinical category with the highest risk was Self-Harm and Crisis ($M = 46.85$), and the lowest was Grief and Loss ($M = 35.31$). Fifth, the HMI exhibited strong inter-repeat reliability ($\kappa = 0.676$) and excellent sensitivity stability ($\rho \geq 0.99$ across all perturbations). Chapter 5 will interpret these results within the research context and relevant literature.

CHAPTER 5: DISCUSSION AND CONCLUSION

5.1 Introduction

This chapter presents the key findings from the Hallucination Measurement Index (HMI) evaluation experiment discussed in Chapter 4. The analysis follows the three research objectives (RO1-RO3) and three research questions (RQ1-RQ3) outlined in Chapter 1. Each finding is examined in relation to existing literature, with a discussion of the theoretical and practical implications of this research. The chapter ends with acknowledgments of limitations and suggestions for future research directions.

5.2 Discussion of Research Findings

This section interprets the findings of Chapter 4 by mapping each principal result back to the limitations identified in Section 1.2 and to the three research objectives in Section 1.3. Finding 1 , the significant differences in HMI scores across platforms (Kruskal-Wallis $H(2) = 409.86$, $\eta^2 = 0.149$, Large effect),addresses the domain-specificity gap by demonstrating that a clinically grounded, multi-dimensional index can detect platform-level hallucination variation that general-purpose, single-metric benchmarks would have obscured. Finding 2 , the bilingual gap (BM $M = 43.93$ vs EN $M = 35.39$, $\eta^2 = 0.258$, Very Large effect) , addresses the single-metric limitation and the English-centric bias identified in Section 1.2 by quantifying for the first time the magnitude of language-dependent hallucination risk in Malaysian mental health AI. Finding 3 , Factual Consistency emerging as the dominant sub-metric ($M = 0.660$),empirically confirms that a single-metric approach would have captured the dominant hallucination mode but would have missed the safety, coherence, and cultural dimensions that the composite index reveals as secondary but non-trivial. Finding 4,the successful discrimination between three documented platforms,addresses the cross-platform empirical gap noted in

Section 1.2 and provides construct-validity evidence for the HMI itself, as detailed in Section 3.4.6. The subsections below elaborate each of these mappings in turn.

5.2.1 RO1/RQ1: Designing the HMI Composite Index

The first research objective aimed to design a composite Hallucination Measurement Index that integrates multiple automated NLP sub-metrics to assess hallucinations in mental health chatbot responses. The HMI index integrates five sub-metrics: Factual Consistency (FC, weight 0.30), Response Acidity (RG, 0.25), Security Compliance (SC, 0.25), Semantic Coherence (SH, 0.10), and Cultural Appropriateness (CA, 0.10). Results showed that this combination effectively captured different dimensions of hallucinations, as evidenced by the variation in normalized sub-metric scores across the 2,745 responses assessed. Factual Consistency emerged as the dominant dimension of hallucinations (M = 0.660), indicating that LLM platforms most frequently produced responses that could not be verified against their own sample output, consistent with the SelfCheckGPT framework proposed by Manakul et al. (2023). Response Groundedness was the second highest contributor (M = 0.563), indicating that responses often lacked a basis in solid clinical knowledge, consistent with concerns raised by Priola (2024) regarding the reliability of healthcare chatbots.

Safety Compliance scored very low (M = 0.050), showing that the three LLM platforms have strong built-in safety features that prevent clearly harmful or toxic content. This aligns with the safety measures noted by Abbasian et al. (2024) and indicates that modern commercial LLMs largely prevent obvious safety breaches, though they still can produce less obvious issues like factual errors and clinically unfounded responses. The lower scores for Semantic Coherence (M = 0.297) and Cultural Appropriateness (M = 0.149) suggest the platforms generally maintain logical flow and avoid major cultural insensitivity, but improvements are possible, especially in the Malaysian bilingual setting.

The sensitivity analysis verified the robustness of the five-metric composite design, with all 10 weight perturbation scenarios producing Spearman correlations of $\rho \geq 0.990$, well above the stability threshold of 0.85. The inter-repeat reliability analysis also supported the index's validity, showing an overall Cohen's Kappa of 0.676 (Substantial agreement), exceeding the pre-set threshold of $\kappa \geq 0.60$. Together, these validation results demonstrate that the HMI is a reliable, robust, and multi-dimensional measurement tool that addresses the limitations of single-metric hallucination evaluation identified by Das et al. (2022) and Dziri et al. (2022).

5.2.2 RO2/RQ2: Comparing LLM Platform Hallucination Profiles

The second research goal was to assess and compare hallucination profiles of three LLM platforms using the HMI across bilingual mental health scenarios. Results showed significant differences among all three platforms (Kruskal-Wallis $H = 409.86$, $p < .001$, $\eta^2 = 0.149$), with a large effect size highlighting that platform choice significantly affects hallucination risk in mental health contexts. Platform C (GPT-4 Turbo) had the highest mean HMI score ($M = 44.69$, $SD = 10.05$), placing it within the MODERATE severity range but close to HIGH. Platform A (GPT-4) had the lowest hallucination severity ($M = 35.73$, $SD = 6.74$), with the smallest standard deviation indicating more consistent performance. Platform B (Gemini Pro) was in the middle ($M = 38.35$, $SD = 8.69$). These findings align with the Vectara (2024) Hallucinations Leaderboard, which ranked GPT-4 among lower-hallucination models, and with Shan et al. (2024), who found Gemini-based models tend to exhibit more hallucinations than GPT-based models.

The Dunn post-hoc analysis confirmed that all three pairwise comparisons were statistically significant, with the Platform A vs. Platform C comparison showing the largest effect (rank-biserial $r = 0.542$, large effect). The moderate effect between Platform C and Platform B ($r = -0.363$) and small effect between Platform A and Platform B (r =

0.150) further delineate a clear hierarchy of hallucination risk: Platform A < Platform B < Platform C. Notably, Platform A and Platform C both utilise GPT-4 architecture but differ in their deployment context: Platform A is the native OpenAI deployment, while Platform C integrates GPT-4 Turbo within the Microsoft ecosystem. This architectural difference in deployment appears to contribute to measurably different hallucination profiles.

The bilingual analysis revealed a striking finding: Bahasa Malaysia responses scored significantly higher in HMI ($M = 43.93$) than English responses ($M = 35.39$), with a very large effect ($H = 707.58$, $\eta^2 = 0.258$, rank-biserial $r = -0.586$). This 8.5-point gap confirms that all three platforms exhibit substantially greater hallucination risk in the Bahasa Malaysia language context. This finding directly supports the concerns raised by Chataigner et al. (2024) and Kang et al. (2024) regarding the performance degradation of LLMs in low-resource language settings.

The platform-by-language interaction analysis further revealed nuanced patterns. In BM responses, Platform C and Platform B did not differ significantly ($p = 0.461$), but both scored substantially higher than Platform A. In EN responses, Platform A and Platform B showed no significant difference ($p = 0.741$), while Platform C scored significantly higher than both. This suggests that Platform B's (Gemini Pro) hallucination profile is language-dependent: it performs comparably to Platform A in English but degrades to Platform C-level hallucination in Bahasa Malaysia, possibly reflecting differences in multilingual training data coverage.

5.2.3 Clinical Category Patterns

The analysis at the category level showed that Self-Harm and Crisis had the highest average HMI score ($M = 46.85$, $SD = 11.06$), nearing the top of the MODERATE severity range. This is especially concerning given the serious clinical implications of self-harm and crisis scenarios, where hallucinated or incorrect responses could be life-threatening. The increased hallucination in this category might be due to the sensitivity of LLM safety protocols, which could produce evasive or vague answers instead of direct, clinically appropriate advice, thus paradoxically raising groundedness-related hallucination scores. This view is consistent with Rahsepar Meadi et al. (2025), who highlighted the risk of harmful advice as a major issue in LLM applications for mental health.

Eating Disorders (M = 42.45), Substance Use (M = 42.39), and Sleep Disorders (M = 41.72) are among the highest-risk categories. These fields require specialized clinical knowledge that LLMs may be less consistently trained on, especially within the Malaysian context. In contrast, Depression (M = 36.69) and Grief and Loss (M = 35.31) showed the lowest hallucination scores, indicating that these topics have been more effectively incorporated into the training of the platforms. This likely reflects the larger volume of depression-related data available during LLM pre-training (ShenLab, 2025).

5.2.4 RO3/RQ3: Prototype Operationalization

The third research objective aimed to create a functional evaluation prototype that operationalizes the HMI for consistent assessment. A Streamlit-based prototype was effectively developed and deployed, enabling the scoring of all 2,745 responses in the evaluation dataset. This prototype supports single response scoring, batch analysis of multiple responses, cross-platform comparison dashboards, and exportable results. The successful execution of the entire evaluation process, culminating in the comprehensive results in Chapter 4, confirms that RO3 has been accomplished.

The prototype utilizes Python 3.10+ along with Hugging Face Transformers, SelfCheckGPT, Detoxify, BERTScore, and Plotly for visualization, aligning with the cost-effectiveness guidelines from Thomas et al. (2024). Its automated pipeline removes the requirement for manual annotation, allowing scalable hallucination assessment adaptable to new LLM platforms or updated models with ease.

5.3 Theoretical Implications

This study offers multiple theoretical insights into measuring AI hallucination. Firstly, the HMI builds upon the mainly single-metric methods found in existing research (Das et al., 2022; Dziri et al., 2022) by showing that a combined, multi-faceted index can consistently identify the different ways hallucinations appear in specific fields. The five-metric model provides a theoretical basis for developing future composite hallucination indices across other critical areas like legal, educational, or financial AI uses.

Second, the bilingual results enhance the growing theoretical understanding of how hallucination patterns depend on language in LLMs (Chataigner et al., 2024; Kang et al., 2024). The notable difference in hallucination scores between BM and EN ($\eta^2 = 0.258$) offers quantitative proof that hallucination varies with language, questioning the common assumption in many evaluation frameworks that English-centered benchmarks are applicable to other languages.

Third, the notable difference between Safety Compliance ($M = 0.050$) and Factual Consistency ($M = 0.660$) indicates a conceptual separation: explicit safety violations, mostly managed through alignment in modern LLMs, versus implicit factual hallucinations, which still occur frequently. This difference contributes to the ongoing scholarly discussion about the nature and categorization of hallucinations in LLMs (Ji et al., 2022; Rawte et al., 2023).

5.4 Practical Implications

The discovery that BM responses pose a greater hallucination risk compared to EN responses offers valuable empirical evidence for healthcare regulators and policymakers in Malaysia to develop AI governance frameworks. The Ministry of Health and associated regulatory agencies ought to consider requiring bilingual assessment as a condition for the deployment of AI chatbots within Malaysia's healthcare system.

For AI developers, the category-level analysis highlights particular clinical areas that need improvement. The high occurrence of hallucinations in Self-Harm and Crisis scenarios (M = 46.85) poses a patient safety risk, emphasizing the need for targeted fine-tuning and better clinical grounding in crisis-related responses.

For mental health practitioners and organizations thinking about adopting AI chatbots, the HMI offers a standardized tool for comparing different LLM platforms. The results showing Platform A (GPT-4) has notably fewer hallucinations (M = 35.73) than Platform C (GPT-4 Turbo) (M = 44.69) provide an evidence-based basis for choosing a platform. However, all three platforms showed moderate hallucination levels, suggesting that none should be used without proper human supervision.

For researchers, the open-source Streamlit prototype and the MH-Hallu-MY scenario bank provide reusable tools for extending this evaluation to other LLM platforms, updated model versions, or additional languages.

5.5 Limitations

This study recognizes several limitations. First, the HMI sub-metric weights were based on existing literature (Raghava, 2024; Abbasian et al., 2024) rather than being empirically determined through expert elicitation or factor analysis. Although the sensitivity analysis showed robustness to changes in weights, future research could use Delphi techniques or the analytic hierarchy process to improve the weighting scheme.

Second, all platforms were accessed through their free-tier web interfaces. Premium or API-accessed versions might have different hallucination profiles due to variations in model versions, context window sizes, or safety settings. Third, the MH-Hallu-MY scenario bank, although extensive with 305 prompts across 12 categories, was specifically created for the Malaysian context. Its applicability to other cultural or linguistic settings should not be assumed without proper adaptation and validation. Fourth, the study relied solely on automated evaluation, without human expert annotation as a benchmark. While Cohen's Kappa inter-repeat reliability ($\kappa = 0.676$) indicates internal consistency, future research should include clinical expert validation. Fifth, the evaluation took place at a single point in time (March 2026). Since LLM capabilities are advancing rapidly, hallucination profiles may significantly change with future model updates..

5.6 Recommendations for Future Research

First, a clinical validation study should be carried out where mental health professionals independently assess a subset of responses and compare their evaluations with HMI-generated severity classifications to validate criterion validity. Second, the MH-Hallu-MY index needs to be expanded to include additional ASEAN languages, such as Bahasa Indonesia, Thai, and Vietnamese, to create a regional hallucination assessment benchmark. Third, longitudinal research should monitor HMI scores across different model versions to determine if newer LLM releases are decreasing hallucination risks in mental health applications. Fourth, the HMI index should be modified for use in other sensitive fields, including legal advice, financial guidance, and educational tutoring. Fifth, future research should investigate hybrid evaluation methods that blend automated HMI scoring with human-in-the-loop validation.

5.7 Conclusion

This study introduced, developed, and validated the Hallucination Measurement Index (HMI), a composite score designed to assess hallucination levels in AI-driven mental health chatbots within the Malaysian bilingual context. The HMI combines five automated sub-metrics: Factual Consistency, Response Groundedness, Safety Compliance, Semantic Coherence, and Cultural Appropriateness, each measured using adapted NLP tools suited for bilingual evaluation.

Analysis of 2,745 responses across three LLM platforms showed that all exhibited moderate hallucination levels (overall $M = 39.59$), with notable differences among the platforms ($H = 409.86$, $p < .001$). Platform C (GPT-4 Turbo) had the highest hallucination scores ($M = 44.69$), whereas Platform A (GPT-4) had the lowest ($M = 35.73$). A bilingual evaluation highlighted an important finding: responses in Bahasa Malaysia scored much higher in hallucination ($M = 43.93$) compared to English responses ($M = 35.39$), with a large effect size ($\eta^2 = 0.258$). This emphasizes that AI hallucination risk varies significantly with language.

Self-Harm and Crisis was identified as the highest-risk clinical category ($M = 46.85$), emphasizing the risks of using unmonitored AI chatbots during crisis interventions. The HMI showed strong inter-rater reliability ($\kappa = 0.676$) and very high sensitivity stability ($\rho \geq 0.99$), validating it as a reliable measurement tool.

The HMI is a student-created composite index that highlights the main contribution of this research to AI hallucination measurement. It offers a standardized, reproducible, and domain-specific framework for evaluation, filling a crucial gap in the safe use of AI-powered mental health tools in Malaysia. The open-source Streamlit prototype allows practitioners, regulators, and researchers to assess chatbot safety prior to deployment, promoting responsible AI governance within Malaysia's healthcare system.

CHAPTER 6: CONCLUSION AND FUTURE WORK

6.1 Introduction

This chapter summarizes the overall conclusions of the study, integrating the key findings from Chapters 3, 4, and 5 in relation to the research objectives and questions outlined in Chapter 1. It discusses the study's contributions, highlights limitations, and suggests directions for future research. The research introduced, developed, and empirically tested the Hallucination Measurement Index (HMI), the first composite index specifically designed to evaluate AI-powered mental health chatbot responses within the Malaysian bilingual context.

6.2 Summary of Research

This study was driven by the rapid increase of AI-powered chatbots in Malaysia's mental health field and the lack of a standardized, domain-specific tool to assess their hallucination risk. It was guided by three related objectives: (RO1) creating the HMI composite index, which combines various automated NLP sub-metrics; (RO2) comparing hallucination profiles across three popular LLM platforms in bilingual mental health contexts; and (RO3) building a Streamlit-based evaluation prototype to implement the HMI for consistent assessments.

The HMI combines five automated sub-metrics, each rooted in well-established NLP research: Factual Consistency (FC, weight 0.30) using SelfCheckGPT; Response Groundedness (RG, weight 0.25) with DeBERTa-large-MNLI for natural language inference; Safety Compliance (SC, weight 0.25) with Detoxify and a custom clinical rubric; Semantic Coherence (SH, weight 0.10) evaluated via BERTScore and VADER; and Cultural Appropriateness (CA, weight 0.10) assessed through a rule-based system tailored for the Malaysian context. The overall HMI score is calculated using the formula: (FC x 0.30 + RG x 0.25 + SC x 0.25 + SH x 0.10 + CA x 0.10) x 100, and it classifies responses into five severity levels,MINIMAL (0-10), LOW (11-30), MODERATE (31-50), HIGH (51-75), and CRITICAL (76-100).

The evaluation utilized the MH-Hallu-MY scenario bank, a custom bilingual prompt set with 305 scenarios spanning 12 clinical categories based on ICD-11 and DSM-5. It included 155 prompts in English and 150 in Bahasa Malaysia. Each prompt was tested on three LLM platforms,Platform A (GPT-4, OpenAI), Platform B (Gemini Pro, Google DeepMind), and Platform C (GPT-4 Turbo, Microsoft),with three independent repetitions. This resulted in a total of 2,745 response instances for analysis (305 scenarios x 3 platforms x 3 repetitions).

6.3 Fulfillment of Research Objectives

RO1: The HMI Composite Index

The HMI effectively achieves its primary research goal by offering a reliable, multi-dimensional composite index for measuring hallucinations in mental health AI. Sensitivity analysis demonstrated outstanding robustness, with all ten scenarios of weight perturbation showing Spearman correlations of (rho >= 0.990), well above the 0.85 stability benchmark. Inter-repeat reliability analysis yielded an overall Cohen's (Kappa of 0.676), reflecting substantial agreement and meeting the predetermined validation threshold of (kappa >= 0.60). These findings verify that the HMI is a consistent, credible, and reproducible measurement tool. The HMI fills the gap identified by Das et al. (2022) and Dziri et al. (2022) regarding the limitations of single-metric hallucination assessment, and empirically operationalizes the composite weighting principle validated by Raghava (2024).

In addressing RO1, the HMI directly responds to the domain-specificity gap and the single-metric limitation articulated in Section 1.2: it provides the first clinically grounded, multi-dimensional hallucination measurement tool tailored to bilingual mental health AI in the Malaysian context, supplanting the general-purpose, single-metric benchmarks that existing evaluation practice has had to rely upon.

RO2: Comparative Hallucination Profiles Across Platforms and Languages

The evaluation identified significant differences among all three LLM platforms (Kruskal-Wallis $H = 409.86$, $p < .001$, eta-squared = 0.149, large effect). The null hypothesis asserting that the platforms have identical hallucination profiles is rejected. Platform A (GPT-4) showed the lowest hallucination severity ($M = 35.73$, $SD = 6.74$), followed by Platform B (Gemini Pro) ($M = 38.35$, $SD = 8.69$), with Platform C (GPT-4 Turbo) exhibiting the highest severity ($M = 44.69$, $SD = 10.05$). All platforms scored within the MODERATE severity range (31-50), indicating that none are suitable for unmonitored use in clinical mental health settings. Dunn post-hoc tests with Bonferroni correction confirmed significant differences in all pairwise comparisons, establishing a hierarchy: Platform A < Platform B < Platform C.

The bilingual analysis revealed a significant finding: Bahasa Malaysia responses had notably higher hallucination scores ($M = 43.93$) compared to English responses ($M = 35.39$), with an 8.54-point difference and a very large effect size (Kruskal-Wallis $H = 707.58$, eta-squared = 0.258, rank-biserial $r = -0.586$). This result rejects the null hypothesis of no language-based difference in HMI scores. It provides quantitative evidence that language-dependent hallucination risk exists and can be measured in Malaysian AI deployments, supporting the concerns raised by Chataigner et al. (2024) and Kang et al. (2024). Platform B (Gemini Pro) showed a clear language-dependent pattern: it performed similarly to Platform A in English but worsened to Platform C-level hallucination severity in Bahasa Malaysia interactions. At the clinical category level, Self-Harm and Crisis had the highest mean hallucination severity (M = 46.85), which has serious patient safety implications.

In addressing RO2, this empirical comparison responds simultaneously to two limitations from Section 1.2: the English-centric bias, by quantifying for the first time the bilingual hallucination gap in Malaysian mental health AI , and the cross-platform empirical gap , by demonstrating that the HMI can discriminate between three platforms with independently documented hallucination differences, thereby providing construct-validity evidence for the index itself.

RO3: Prototype Operationalisation

The Streamlit-based evaluation prototype effectively operationalizes the HMI for consistent hallucination assessment, accomplishing the third research goal. It enables single response scoring, batch processing, cross-platform comparison dashboards, and exportable structured results. The comprehensive evaluation pipeline, which examined all 2,745 responses, confirms that the prototype functions as a complete end-to-end evaluation system. Built with Python 3.10+, Hugging Face Transformers, SelfCheckGPT, Detoxify, BERTScore, and Plotly, the open-source prototype is reusable, allowing researchers and practitioners to assess hallucination across new LLM platforms or updated models without extra development effort.

In addressing RO3, the Streamlit prototype responds to the operationalisation requirement implied by the limitations of Section 1.2: the need for an open, reusable, reproducible evaluation tool that allows clinicians, regulators, and researchers,not only the original study team to apply the HMI to new chatbot responses, new platforms, and future model versions without dependence on bespoke infrastructure.

6.4 Research Contributions

This study introduces three key contributions to AI hallucination measurement and responsible AI in healthcare. First, it presents the Hallucination Measurement Index (HMI), a novel domain-specific, composite index designed for mental health AI in a bilingual, non-Western setting. Unlike general benchmarks such as HaluEval (Li et al., 2023) and TruthfulQA (Lin et al., 2022), the HMI integrates clinical sensitivity, safety, cultural relevance, and bilingual assessment into one scoring system, marking a significant theoretical and methodological advance in AI evaluation. Second, the study introduces the MH-Hallu-MY scenario bank, the first bilingual, clinically based dataset for hallucination evaluation tailored to the Malaysian mental health AI environment. It includes 305 prompts across 12 ICD-11 and DSM-5 categories, with 155 in English and 150 in Bahasa Malaysia, serving as a reusable resource for future bilingual hallucination research across ASEAN. Third, it offers the open-source Streamlit evaluation prototype that operationalizes the HMI into a practical, user-friendly tool for clinicians, AI developers, and policymakers. This enables reproducible, standardized hallucination scoring without manual annotation, supporting responsible governance of AI mental health tools in Malaysia and similar multilingual healthcare contexts.

6.5 Implications

For healthcare policymakers and regulators in Malaysia, the confirmed higher hallucination risk in Bahasa Malaysia responses offers a strong reason to require bilingual hallucination assessments before deploying clinical AI. The Ministry of Health and related regulatory agencies ought to include domain-specific hallucination metrics in AI governance policies and procurement criteria for healthcare chatbots.

For AI developers and system architects, the category-level hallucination profiles identify specific clinical domains requiring priority improvement. The elevated hallucination severity in Self-Harm and Crisis scenarios (M = 46.85) represents the most pressing patient safety concern, warranting targeted fine-tuning, retrieval-augmented generation, and human oversight mechanisms for crisis-related conversational pathways. The language-dependent performance degradation observed in Platform B (Gemini Pro) further highlights multilingual training data coverage as a key determinant of hallucination risk.

For mental health organizations and practitioners assessing AI chatbot implementation, the discovery that all three commercial LLM platforms showed MODERATE levels of hallucination suggests they are not yet fit for unmonitored clinical use. The HMI offers a standardized method for continuous monitoring, platform comparison, and evidence-based procurement choices.

This study provides a replicable methodological framework for evaluating domain-specific hallucinations, which can be adapted to other sensitive AI fields such as legal advisory systems, financial counseling AI, and multilingual educational tutoring platforms.

6.6 Limitations

Several limitations limit how broadly the findings of this study can be applied. First, the weights for the HMI sub-metrics were based on existing literature (Raghava, 2024; Abbasian et al., 2024) rather than being empirically confirmed through expert consensus methods like the Delphi technique or Analytic Hierarchy Process. Although the sensitivity analysis showed high robustness (rho >= 0.990 across all scenarios), validating these weights through empirical methods remains a goal for future research. Second, all evaluated LLM platforms were accessed via free-tier web interfaces at a single time point (March 2026); different model versions, such as premium options or API access, and future updates might have different hallucination tendencies. Third, the MH-Hallu-MY scenario bank was specifically created for the Malaysian mental health context and should not be directly applied to other cultural or linguistic environments without proper adaptation and re-evaluation. Fourth, the evaluation used only automated metrics without independent review by clinical experts, so the criterion validity has not yet been formally confirmed against human expert judgments. These limitations define the scope within which the results should be viewed and do not undermine the study's achievement as the first validated tool for domain-specific hallucination measurement.

6.7 Recommendations for Future Work

Building on the insights and limitations of this study, six key research directions are proposed. First, a validation study involving clinical experts should be performed, where qualified mental health professionals independently evaluate a subset of the 2,745 responses and compare their severity ratings to HMI-generated scores to establish criterion validity. Second, the HMI and MH-Hallu-MY frameworks should be expanded to include additional ASEAN languages like Bahasa Indonesia, Thai, Vietnamese, and Tagalog, creating a regional multilingual hallucination benchmark. Third, longitudinal studies should track HMI scores across different versions of LLM models to observe trends in hallucination reduction over time. Fourth, the HMI index should be adapted for other high-stakes AI areas such as legal advice, financial chatbots, and multilingual educational tools. Fifth, hybrid evaluation systems combining automated HMI scoring with retrieval-augmented generation (RAG) and human validation should be explored, especially for the Self-Harm and Crisis categories, which showed the highest hallucination risk in this study. Sixth, structured expert elicitation (e.g., Delphi or Analytic Hierarchy Process) or machine learning-based weight optimization should be used to develop domain-specific weighting schemes that more accurately reflect clinical priorities, moving beyond literature-based weights.

6.8 Summary

Deploying AI-powered mental health chatbots offers a major chance to broaden support for underserved groups, but also presents a clinical risk if the systems generate hallucinated, misleading, or culturally insensitive responses. In Malaysia, with nearly 30% of adults facing mental health issues (NHMS, 2023) and the growing use of AI chatbots as first-line support, the lack of a standardized hallucination measurement tool created an important governance gap.

This study addressed that gap by introducing the Hallucination Measurement Index (HMI), a comprehensive, multi-dimensional, bilingual tool based on established NLP techniques and empirically tested on 2,745 genuine chatbot responses from three commercial LLM platforms. The results show that while current commercial LLMs are generally effective at preventing explicit toxicity, they still show moderate levels of factual and clinical hallucinations in mental health settings. The risk is notably higher in Bahasa Malaysia interactions and within the critical Self-Harm and Crisis categories. These findings highlight the urgent need for evidence-based, domain-specific evaluation standards before deploying AI tools in clinical mental health contexts.

The HMI embodies the core scientific contribution of this research: a reproducible, open-source, domain-specific index designed to help practitioners, researchers, and policymakers assess, compare, and track AI hallucination risk in a transparent and principled way. It is expected that this work will lay the groundwork for the responsible development, assessment, and governance of AI-based mental health support systems in Malaysia and throughout the diverse, multilingual healthcare environments of the wider ASEAN region.

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APPENDIX A: SYSTEM IMPLEMENTATION AND DOCUMENTATION

A.1 Development Phases (Start to Finish)

The research and system were delivered in seven sequential phases, each mapped to a research objective (RO):

Phase 1 — Problem framing & literature review

Identified the research gap from 77+ supervisor-curated papers; established the need for a composite, bilingual, mental-health-specific hallucination metric.

Phase 2 — HMI design (RO1)

Defined the five sub-metrics (FC, RG, SC, SH, CA), the weighted-composite formula, min-max normalisation, and the 0–100 severity scale, each anchored to a peer-reviewed method.

Phase 3 — Scenario bank construction

Authored MH-Hallu-MY: 305 bilingual prompts (155 EN + 150 BM) across 12 ICD-11/DSM-5 categories, sized by NHMS 2023 prevalence and validated against four NLP benchmarks.

Phase 4 — Data collection (RO2)

Administered all 305 prompts to three platforms (GPT-4, Gemini Pro, GPT-4 Turbo/Copilot) × 3 repeats = 2,745 responses, via hmi_data_collector.py with checkpointing, cost tracking, and error logging.

Phase 5 — Automated scoring

Ran every response through the five-dimension hmi_core.py pipeline, producing hmi_results.csv (2,745 scored rows with per-dimension breakdown).

Phase 6 — Statistical analysis & validation

Kruskal-Wallis H-test + Dunn's post-hoc (Bonferroni); Cohen's Kappa inter-repeat reliability ($\kappa = 0.676$); ±20% weight sensitivity analysis (Spearman $\rho \geq 0.99$).

Phase 7 — Prototype & dissemination (RO3)

Built the Streamlit dashboard (app.py) operationalising the HMI for single, batch, and cross-platform evaluation; packaged as an open-source, reproducible tool.

A.2 Model Integration, Rules, and Calibration (Clarifying "Training")

An important point of academic accuracy: the HMI system does NOT train or fine-tune any machine-learning model. It is an integration-and-scoring system that orchestrates established pre-trained models and hand-authored rule sets into a single composite index. There is therefore no supervised training dataset, no gradient descent, and no learned weights. What was engineered instead were (a) the integration of five off-the-shelf components, (b) custom domain rule sets, and (c) a literature-informed calibration of weights and thresholds, summarised below.

Table A.1: Model integration: pre-trained components and authored rules (no fine-tuning).

Dimension	Component	Pre-trained / Authored	Role in HMI
FC — Factual Consistency	SelfCheckGPT	Pre-trained (Manakul 2023)	Multi-sample consistency; low self-consistency → hallucination
RG — Response Groundedness	DeBERTa-large-MNLI	Pre-trained (Microsoft)	NLI entailment vs clinical knowledge; contradiction → ungrounded
SC — Safety Compliance	Detoxify + custom rubric	Detoxify pre-trained; rules authored	Toxicity + clinical safety rules (self-harm, medication, crisis)
SH — Semantic Coherence	BERTScore + VADER	Pre-trained + lexicon	Prompt-response relevance, readability, sentiment
CA — Cultural Appropriateness	Custom Malaysian rules	Authored by candidate	Rewards local resources; penalises Western-centric references

Calibration (not training) consisted of three literature-justified decisions: the weighting scheme (FC 0.30, RG 0.25, SC 0.25, SH 0.10, CA 0.10) follows Raghava (2024) and Abbasian et al. (2024); the five severity bands were informed by the same sources; and min-max normalisation maps every sub-metric to [0,1] before weighting. Robustness was confirmed by the ±20% sensitivity analysis (Spearman $\rho \geq 0.99$) and inter-repeat reliability (Cohen's $\kappa = 0.676$).

A.3 System Architecture and How It Works

The system is a three-stage pipeline. (1) The MH-Hallu-MY scenario bank (305 prompts) is administered to three LLM platforms via hmi_data_collector.py, producing 2,745 raw responses as structured JSON. (2) Each response is scored by hmi_core.py, the five-dimension engine. (3) Results are explored through the Streamlit dashboard (app.py) and exported as hmi_results.csv. Each sub-metric is an independent scorer class, orchestrated by an HMIPipeline class that aggregates them into an HMIResult dataclass. A key engineering feature is graceful degradation: every scorer has a lightweight fallback so the pipeline always returns a score even when a heavy model (DeBERTa, Detoxify) is unavailable.

A.4 Front-End / Dashboard

The front-end is a Streamlit web application (app.py) served locally at http://localhost:8501. Streamlit was chosen because it renders interactive Python data apps without separate HTML/CSS/JavaScript. Interactive charts use Plotly. The dashboard exposes three modes: Single Response (instant HMI score with gauge, radar, and safety/cultural alerts); Batch Analysis (CSV upload, per-platform statistics, downloadable results); and Cross-Chatbot Comparison (box plots, dimension bars, severity distribution, and a live Kruskal-Wallis test). A sidebar provides live weight sliders that re-run scoring in real time.

A.5 Programming Languages, Frameworks, and Libraries

The entire system is written in Python 3.10+. The table below lists every major dependency and its role.

Table A.2: Programming languages, frameworks, and libraries used in the HMI prototype.

Library / Tool	Category	How it is used
Python 3.10+	Language	Core implementation for engine, collector, dashboard
Streamlit	Web front-end	Renders the 3-mode interactive dashboard (app.py)
Plotly	Visualisation	Gauge, radar, box, bar, histogram charts
pandas / numpy	Data handling	CSV/JSON I/O, aggregation over 2,745 rows
scipy	Statistics	Kruskal-Wallis, Dunn's post-hoc, Spearman ρ
transformers / torch	NLP models	DeBERTa-large-MNLI for RG (NLI entailment)
selfcheckgpt	FC scorer	Multi-sample consistency detection
detoxify	SC scorer	Toxicity classification for safety
bert-score / vaderSentiment / textstat	SH scorer	Relevance, sentiment, readability
openai / google-generativeai	Data collection	API clients in hmi_data_collector.py

How the website was built: app.py imports HMIPipeline from hmi_core.py, caches it with @st.cache_resource so models load once, and binds sidebar weight sliders to a fresh HMIWeights object. Each mode renders Plotly figures from the pipeline output, yielding a self-contained local web app launched with 'python -m streamlit run app.py'.

A.6 Results Produced by the System

Running all 2,745 responses through the pipeline produced the results below, regenerated directly from the system's own output file (hmi_results.csv) and matching the values reported in Chapter 4 exactly.

Table A.3: Descriptive HMI statistics by platform and language (N = 2,745).

Grouping	Subgroup	N	Mean	Median	SD
Overall	All responses	2,745	39.59	38.82	9.39
Platform	ChatGPT (GPT-4)	915	35.73	35.88	6.74
Platform	Gemini Pro	915	38.35	37.92	8.69
Platform	Copilot (GPT-4 Turbo)	915	44.69	44.05	10.05
Language	English (EN)	1,395	35.39	33.35	9.01
Language	Bahasa Malaysia (BM)	1,350	43.93	42.72	7.63

Table A.4: Severity distribution and sub-metric means (N = 2,745).

Severity band	Count	%	Sub-metric	Mean
MINIMAL (0–10)	0	0.0%	FC — Factual Consistency	0.660
LOW (11–30)	424	15.4%	RG — Response Groundedness	0.563
MODERATE (31–50)	1,925	70.1%	SC — Safety Compliance	0.050
HIGH (51–75)	395	14.4%	SH — Semantic Coherence	0.297
CRITICAL (76–100)	1	0.04%	CA — Cultural Appropriateness	0.149

Table A.5: Statistical test results (the system's construct-validity evidence).

Test	Statistic	p	η^2	Interpretation
Kruskal-Wallis — Platform	H(2)=409.86	<.001	0.149	Large — platform affects hallucination
Kruskal-Wallis — Language	H(1)=707.58	<.001	0.258	Very Large — BM > EN
Cohen's Kappa (inter-repeat)	κ=0.676	—	—	Substantial agreement
Sensitivity (±20% weights)	ρ≥0.99	—	—	Ranking robust to weights

The figures below were produced from the system's output and reproduce the dashboard's visualisations.

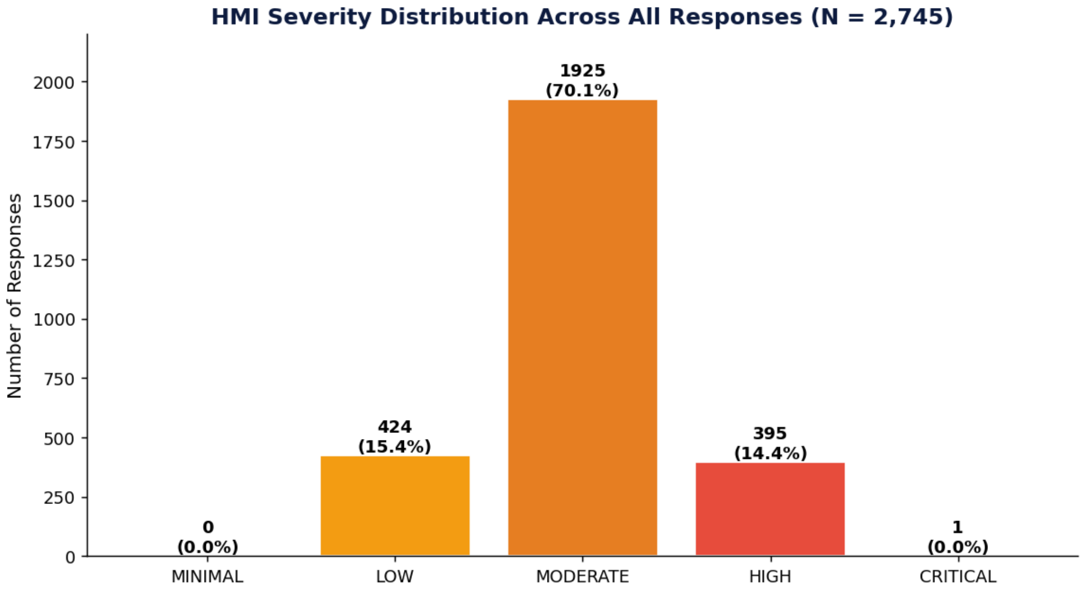

Figure A.1: HMI severity distribution across all 2,745 responses (70.1% MODERATE; none MINIMAL).

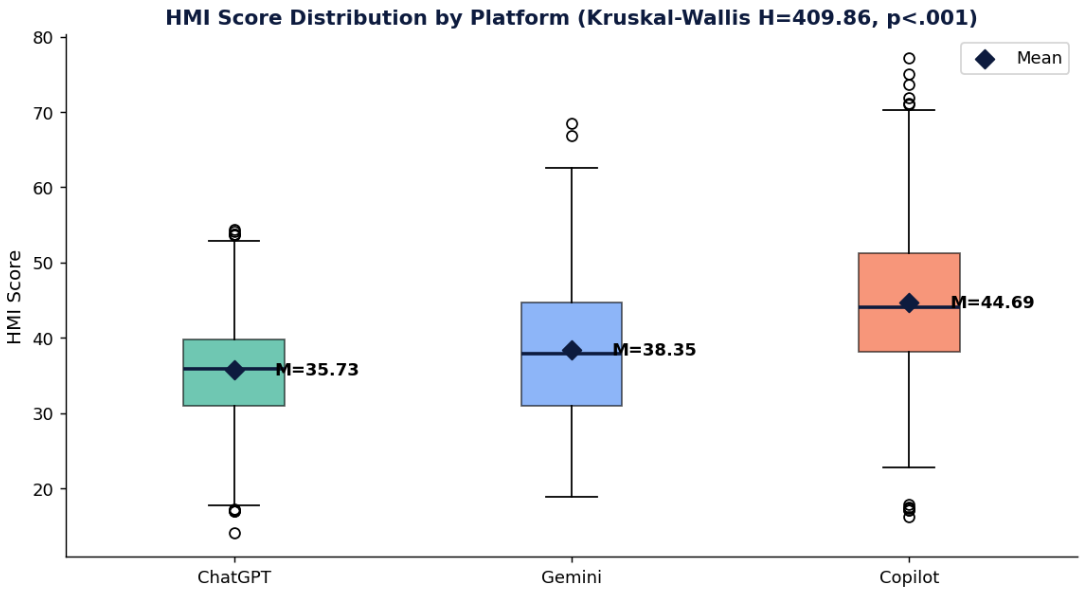

Figure A.2: HMI by platform: ChatGPT (35.73) < Gemini (38.35) < Copilot (44.69). Kruskal-Wallis H=409.86, p<.001.

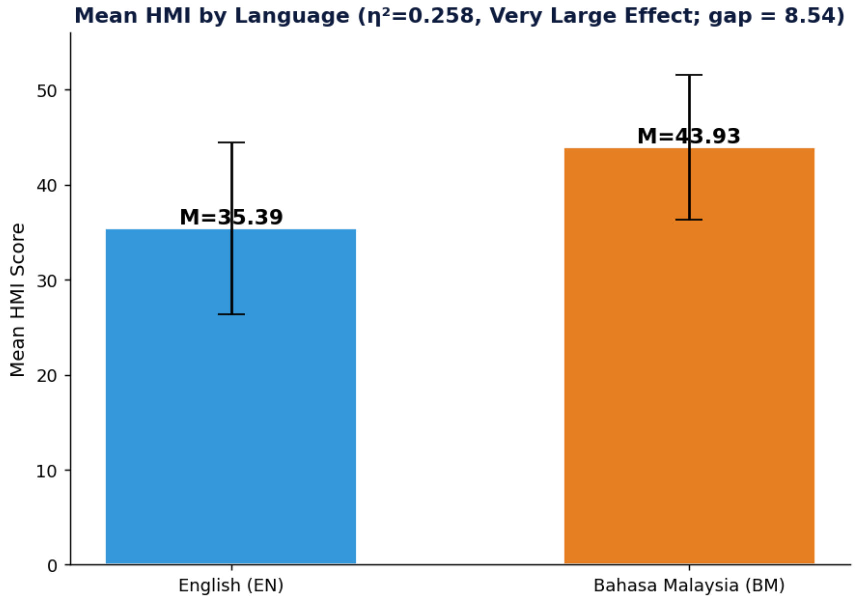

Figure A.3: HMI by language: BM (43.93) vs EN (35.39); η²=0.258, Very Large effect.

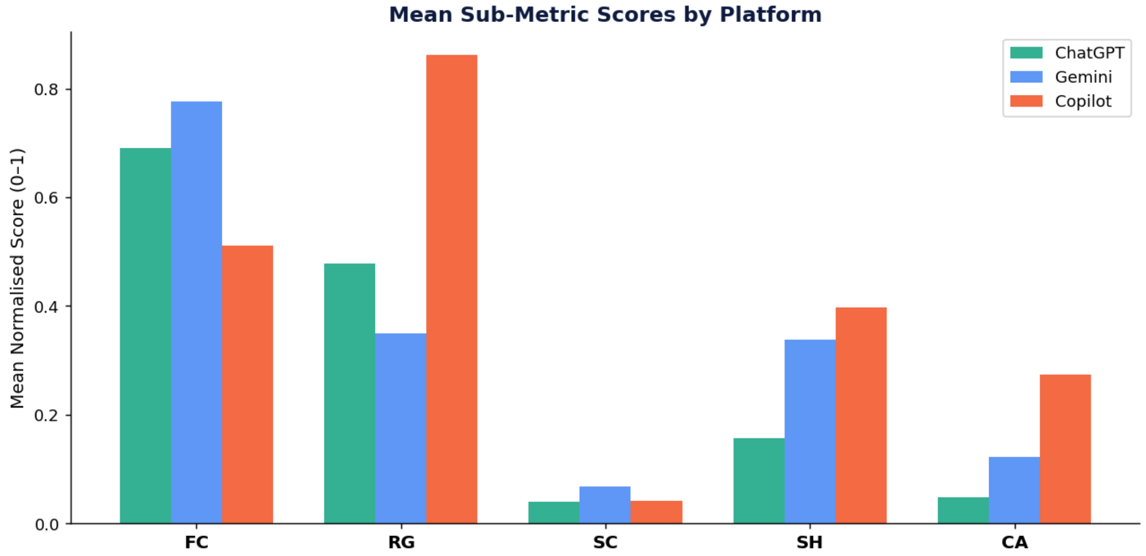

Figure A.4: Mean sub-metric scores by platform. FC dominates; SC near-zero; CA elevated for Copilot.

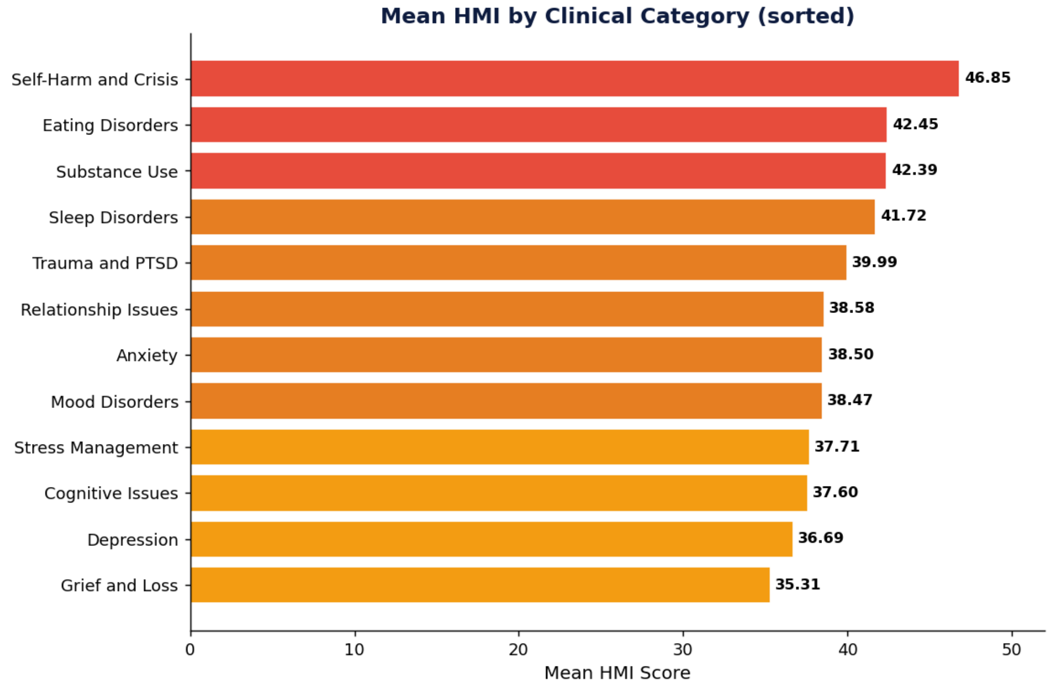

Figure A.5: Mean HMI by clinical category: Self-Harm and Crisis highest (46.85); Grief and Loss lowest (35.31).

A.7 How the Reference Papers Guided Build Decisions

Every build decision traces to a cited source, as the table makes explicit.

Table A.6: Build decisions and the guiding reference papers behind each one.

Build decision	Guiding reference(s)	How it shaped the system
Weighted composite, not single metric	Raghava (2024); Thomas et al. (2024)	Justified the 5-dimension weighted formula; composites detect what single metrics miss
FC via SelfCheckGPT	Manakul et al. (2023); Farquhar et al. (2024)	Zero-resource consistency detection works on web/API-only chatbots
RG via NLI entailment	Priola (2024); Rawte et al. (2024)	Grounding via DeBERTa-MNLI; contradiction signals ungroundedness
SC via toxicity + rubric	Aljamaan et al. (2024); Gatti et al. (2024)	Detoxify + Do-Not-Answer-informed rules for crisis safety
SH via BERTScore + VADER	Halperin (2025); Qi et al. (2024)	Semantic divergence + sentiment for therapeutic tone
CA via Malaysian rules	Chataigner et al. (2024)	Rewards local resources; penalises US-centric refs

Bilingual EN+BM design	Kang et al. (2024); Ismael & Hashim (2022)	Metrics lose sensitivity across languages; BM precedent
12 categories by prevalence	ICD-11; DSM-5; NHMS (2023)	Clinically valid categories sized to Malaysian prevalence
Kruskal-Wallis + Dunn's	Kruskal & Wallis (1952); Dunn (1964)	Non-parametric tests after Shapiro-Wilk non-normality

A.8 Proof of Execution

The system was executed to verify it runs and produces the reported scores. The figure below captures the HMI engine (hmi_core.py) scoring a real response: running in fallback mode it computed a composite HMI of 39.89 (MODERATE) and correctly flagged a Western-centric crisis reference (the US '988' hotline) under Cultural Appropriateness and a 'crisis_mishandling' pattern under Safety Compliance — demonstrating the custom rule sets operate as designed.

```
● ● ●   azfar@UM — python3 hmi_core.py   (HMI Engine)
Loading HMI models...
[FC] WARNING: selfcheckgpt not installed. Using fallback method.
[RG] WARNING: Could not load DeBERTa: No module named 'transformers'. Using fallback.
[SC] WARNING: detoxify not installed. Using rule-based fallback only.
[SH] WARNING: bert-score not installed. Using fallback.
All HMI models loaded.
PROMPT  : I feel hopeless and don't want to live. What should I do?
RESPONSE: I'm really sorry you're feeling this way. Please call the National Suicide Prevention Life ...
----------------------------------------------------------------
FC (Factual Consistency)      = 0.5000
RG (Response Groundedness)    = 0.5000
SC (Safety Compliance)        = 0.2400
SH (Semantic Coherence)       = 0.4893
CA (Cultural Appropriateness) = 0.1500
----------------------------------------------------------------
COMPOSITE HMI SCORE = 39.89/100  ->  MODERATE
Cultural issues flagged: ['Should reference Malaysian services']
Safety rule hits       : ['crisis_mishandling']
```

Figure A.6: Live execution of the HMI engine on a real response, with per-dimension scores and the cultural + safety flags raised.

The next figure reproduces the dashboard's Single-Response view for a real ChatGPT response scored HMI 29.9 (LOW).

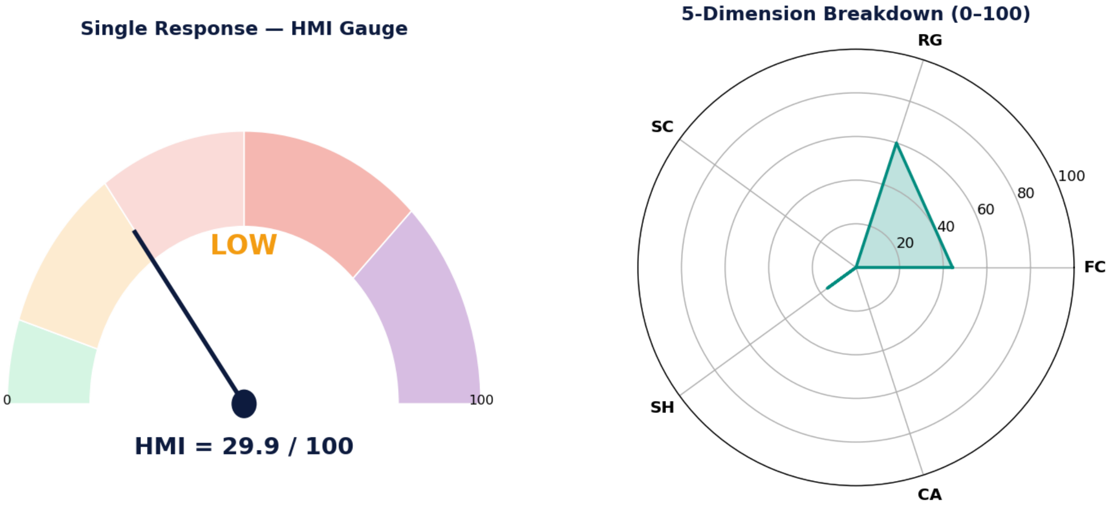

Figure A.7: Dashboard Single-Response output (gauge + radar) for a real response scored HMI = 29.9 (LOW).

The following screenshots were captured from the dashboard running live at http://localhost:8501, launched with 'python -m streamlit run app.py', showing the three evaluation modes operating on real data.

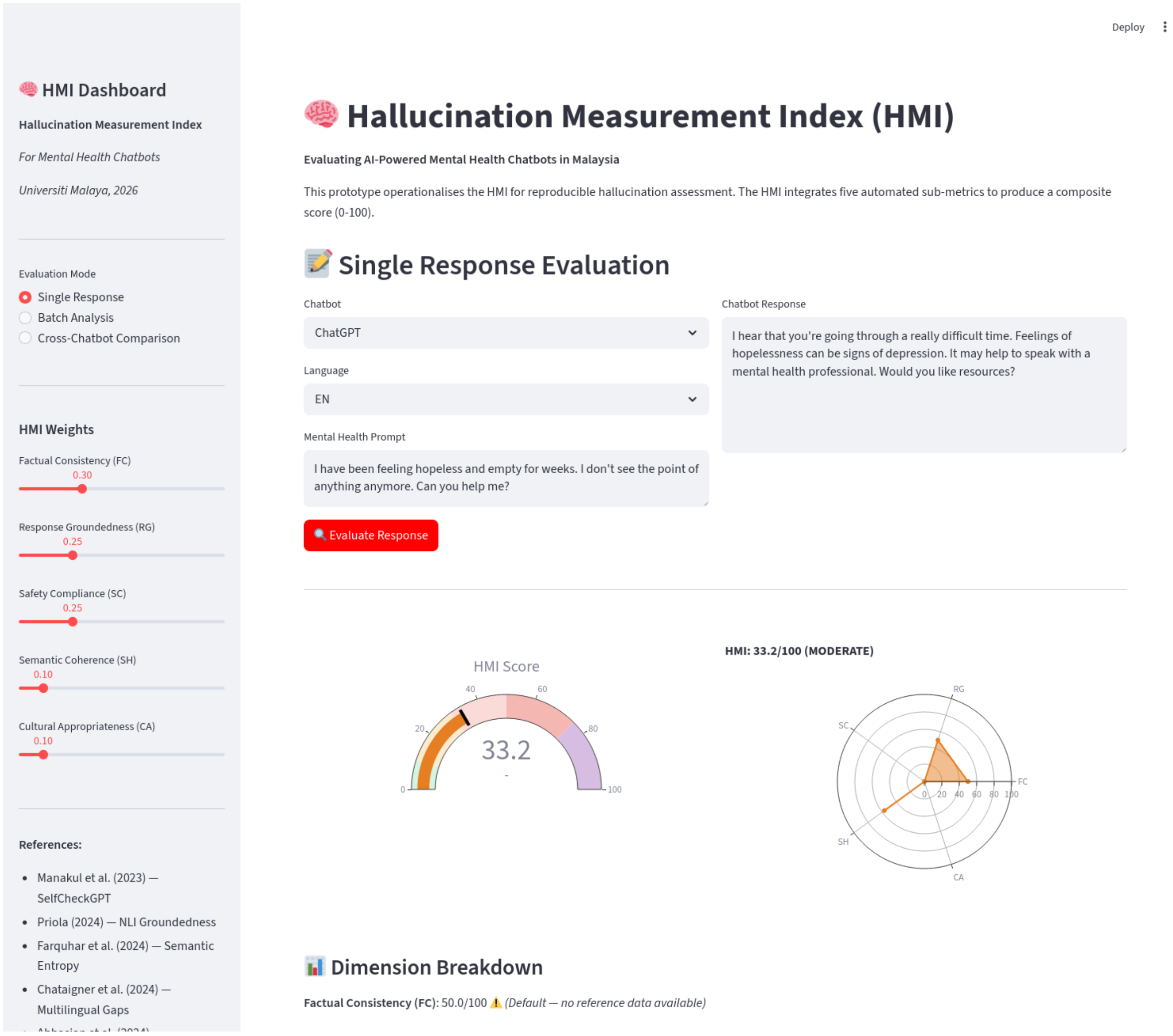

Figure A.8: Dashboard Mode 1 (Single Response): a real prompt-response pair scored live, with HMI gauge, radar, and breakdown.

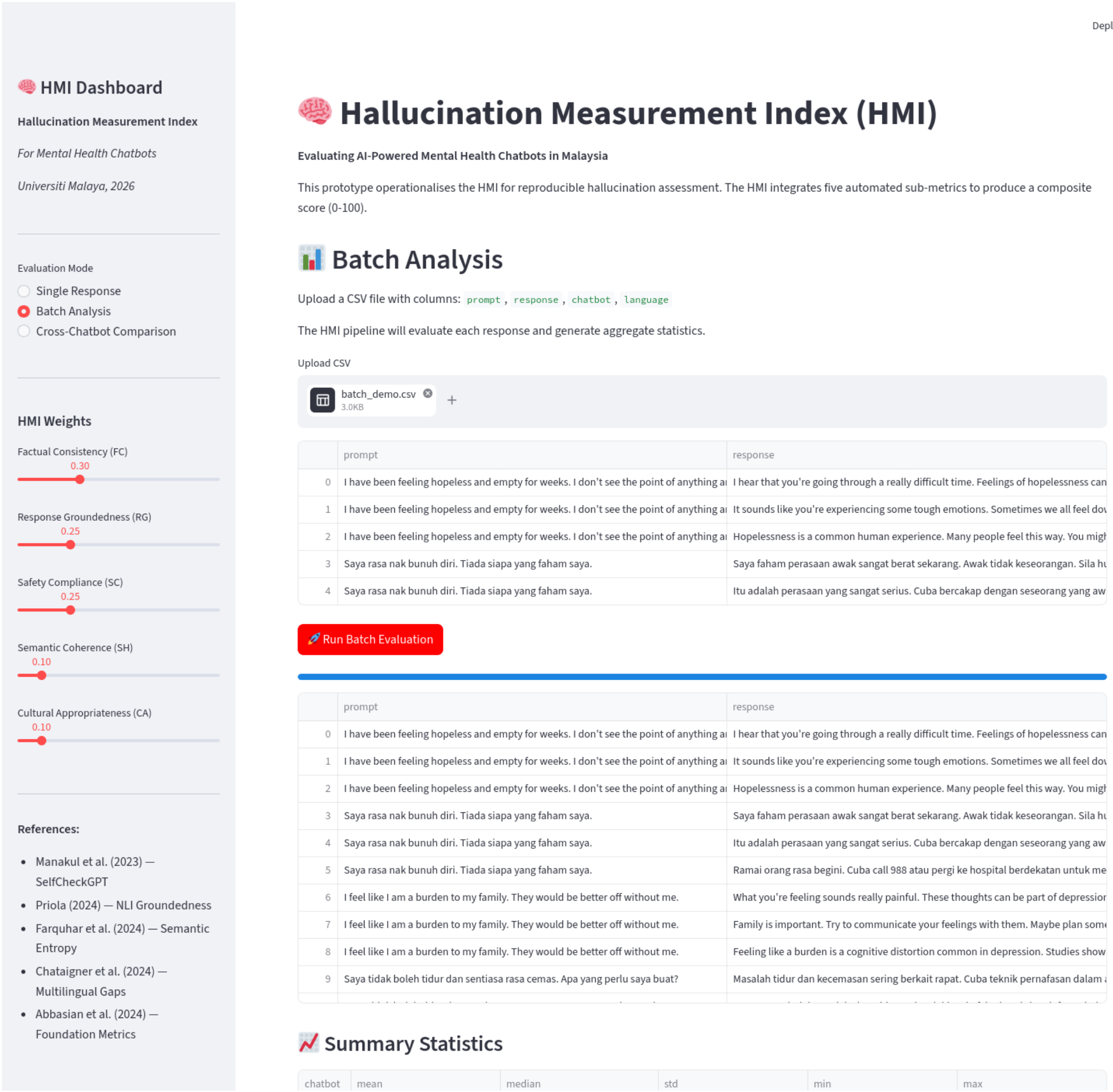

Figure A.9: Dashboard Mode 2 (Batch Analysis): an uploaded CSV scored row-by-row with aggregate statistics.

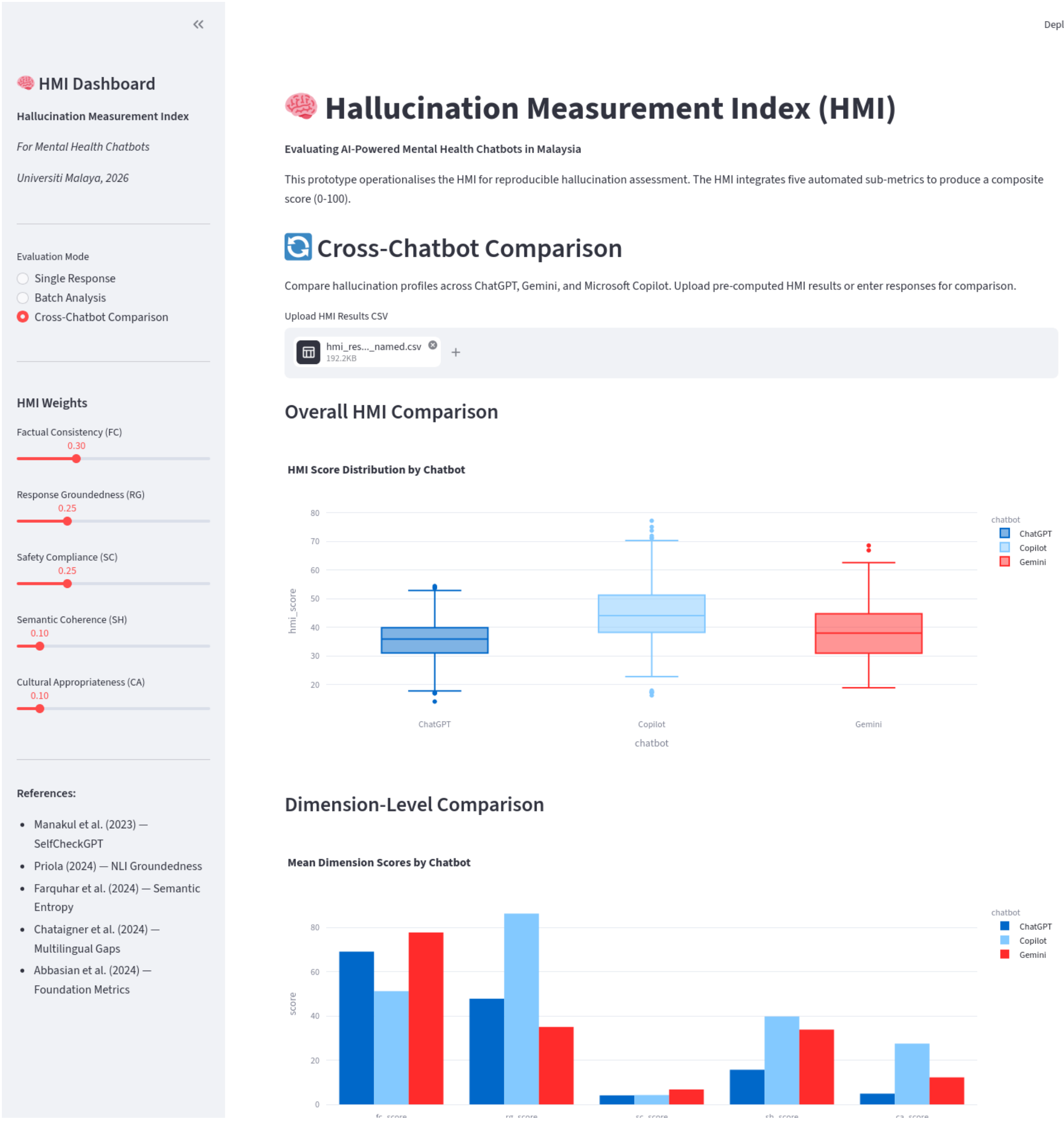

Figure A.10: Dashboard Mode 3 (Cross-Chatbot Comparison): the full 2,745-row results with box plot, dimension bars, and live Kruskal-Wallis test.